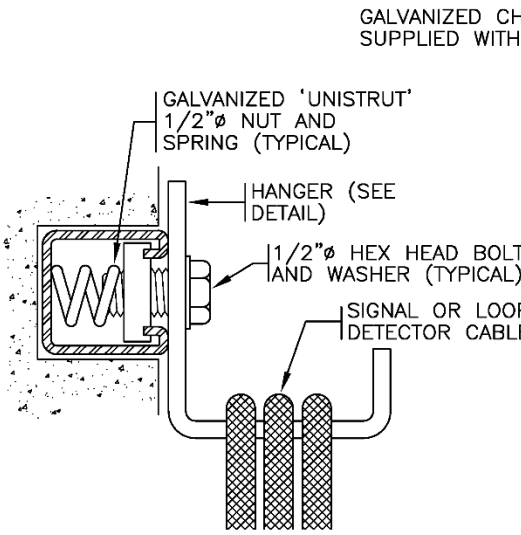
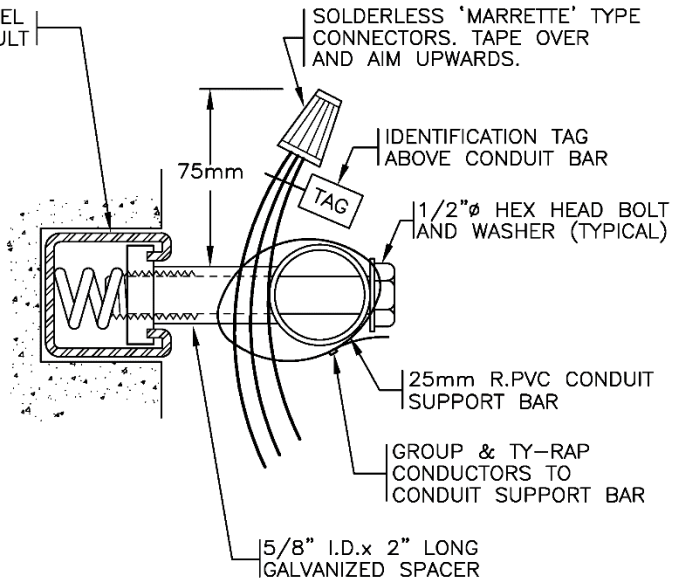
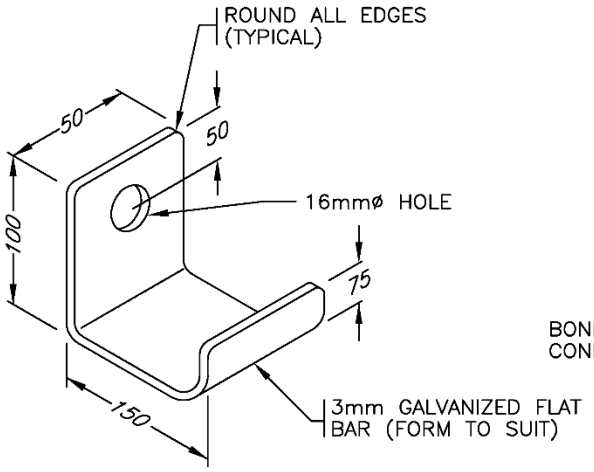


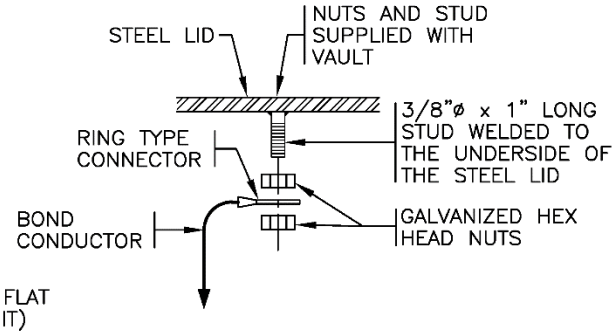
SECTION A–A



SECTION B–B



HANGER



BONDING DETAIL

NOTES

1. SEE DRAWING SP635–2.5.4 FOR ADDITIONAL NOTES & DETAILS.
2. ALL BOLTS & WASHERS SHALL BE GALVANIZED.

NOT TO SCALE

Ministry of Transportation and Infrastructure
BRITISH COLUMBIA

No.	Revision	Date
F		
E		
D		
C		
B	GENERAL REVISIONS	APR 17
A	GENERAL REVISIONS	AUG 94

WIRING INSIDE CONCRETE VAULTS

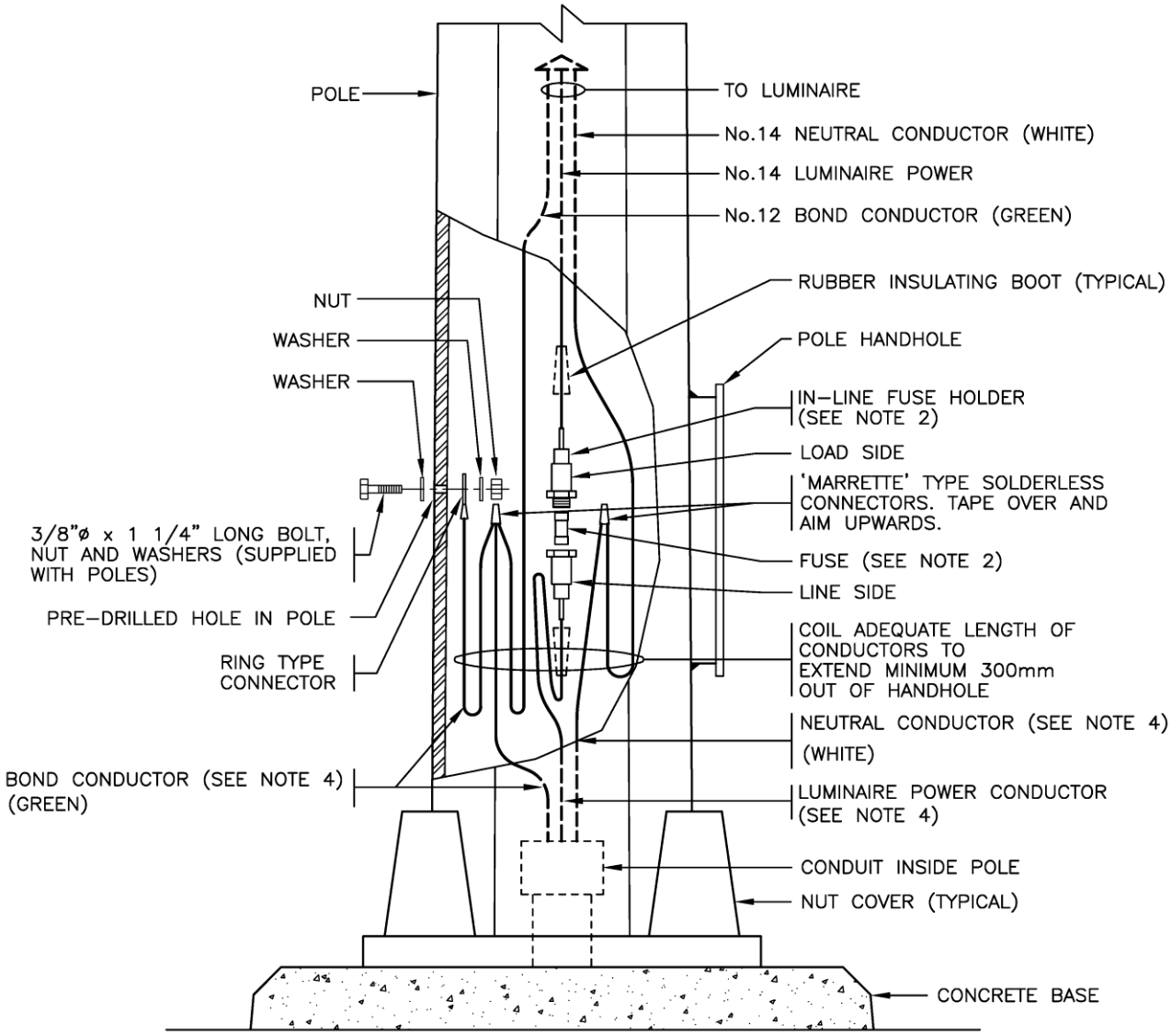
Date Approved

30/09/93 E.L. **(Signature on File)**

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.5.5



ELEVATION

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. IN–LINE FUSE HOLDER SHALL BE A TRON HEB–AA WEATHERPROOF FUSE HOLDER OR A GOULD SHAWMUT GEB–11–11 C/W 5A GOULD SHAWMUT ATM OR BUSS KTK FUSE (347V) OR A 10A GOULD SHAWMUT OTM OR BUSS BAN–10 FUSE (120V) AND 2 'L' TYPE INSULATING BOOTS.
3. FUSE INSTALLATION IN JUNCTION BOXES SIMILAR.
4. SIZE AS PER CEC REQUIREMENTS.

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

NOT TO SCALE

No.	Revision	Date
F	NOTE 4 ADDED	JUN 22
E	GENERAL REVISIONS	APR 20
D	GENERAL REVISIONS	APR 17
C	FUSEHOLDER TYPE REVISED	AUG 99
B	NOTES REVISED	AUG 95
A	GENERAL REVISIONS	AUG 94

LUMINAIRE WIRING IN POLE HANDHOLE

Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.5.6

I.D. TAG DESIGNATIONS	CONDUCTOR COLOUR CODING				
ADV. WARN. FLASH. No.X	ADVANCE WARNING FLASHERS	No. 1	No. 2	No. 3	No. 4
		YELLOW	YELLOW (RD TT)	YELLOW (OR TT)	YELLOW (BL TT)
		BROWN	BROWN	BROWN	BROWN
FLASH. BEACON	FLASHING BEACON	FLASHER CONTROL TO BASE OF POLE : BLACK		BASE OF POLE TO SIGNAL HEADS : RED YELLOW	
ISLAND FLASH.	ISLAND FLASHER	ORANGE			
CONT. PWR.	CONTROLLER POWER	BLACK			
RAIL. PRE-EMPT.	RAILWAY PRE-EMPTION	2C No. 18 SHIELDED CABLE WHITE/BLACK		* SEE NOTE 3	
EMERG. PRE-EMPT. No.X	EMERG. PRE-EMPTION	2C No. 18 SHIELDED CABLE WHITE/BLACK		* SEE NOTE 3	
TEL	TELEPHONE	2C No. 18 SHIELDED CABLE WHITE/BLACK		* SEE NOTE 3	
LOOP No.X	DETECTOR LOOPS	2C No. 18 SHIELDED CABLE WHITE/BLACK		* SEE NOTE 3	
EMERG. IND. LIGHTS	EMERG. INDICATION LIGHTS	BLUE LIGHT	RED (WHITE TT)		
		WHITE LIGHT	RED		
LUM. CCTS. (A, B, ETC.)	LUMINAIRE CIRCUITS	1ø CIRCUITS	A and C	RED	
			B and D	BLACK	
		3ø CIRCUITS	A,D and G	RED	
			B,E and H	BLACK	
			C,F and I	BLUE	
P.E.C.	PHOTOELECTRIC CELL	SWITCH LEG	RED		
		POWER	BLACK		
——	NEUTRAL	WHITE			
——	GROUND	GREEN			

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. TT DENOTES TAPE TRACER (WH – WHITE, BL – BLUE, OR – ORANGE, RD – RED, BR – BROWN)
3. EACH SHIELDED CABLE SHALL BE IDENTIFIED AT BOTH ENDS. IDENTIFICATION SHALL BE MADE USING TY-RAP IDENTIFICATION TAGS (T&B TY5532 OR APPROVED ALTERNATE) WITH THE LOOP NUMBER OR PRE-EMPTION AND PHASE TYPE CLEARLY MARKED USING A BLACK INDELIBLE MARKING PEN.
4. SEE DRAWING SP635-2.8.6 FOR DETECTOR LOOP CONDUCTOR COLOUR CODING.
5. SEE DRAWING SP635-2.5.8 FOR TRAFFIC SIGNAL COLOUR CODING.
6. CONDUCTORS SHALL BE IDENTIFIED IN ALL J.B'S, TRAFFIC CONTROLLERS AND ALL ACCESS POINTS. IDENTIFICATION SHALL BE MADE USING TY-RAP IDENTIFICATION TAGS (T&B TY5532 OR APPROVED ALTERNATE) INDICATING SIGNAL PHASES OR PEDESTRIAN PHASES AS NOTED ON DRAWING SP635-2.5.8. TAGS SHALL BE CLEARLY MARKED WITH A BLACK INDELIBLE PEN.
7. 'X' DENOTES ASSIGNMENT NUMBERS FOR ADVANCED WARNING FLASHERS, FIRE PRE-EMPTIONS, OR LOOPS.

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C	GENERAL REVISIONS	APR 20
B	GENERAL REVISIONS	APR 17
A	GENERAL REVISIONS	AUG 94

SINGLE CONDUCTOR COLOUR CODING

Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-2.5.7

CONDUCTOR COLOUR CODING

TRAFFIC SIGNAL HEAD	SIGNAL SECTION	1	2	3 or x	4 or y
A	RED	RED	RED	RED (BL TT)	RED (BL TT)
	YELLOW	YELLOW	YELLOW (RD TT)	YELLOW	YELLOW (RD TT)
	GREEN	BLUE	BLUE	BLUE	BLUE
	YELLOW ARROW	ORANGE (WH TT)	ORANGE (WH TT)	ORANGE (WH TT)	ORANGE (WH TT)
	GREEN ARROW	BLUE (WH TT)	BLUE (WH TT)	BLUE (WH TT)	BLUE (WH TT)
B	RED	RED	RED	RED (BL TT)	RED (BL TT)
	YELLOW	BROWN	BROWN (OR TT)	BROWN	BROWN (OR TT)
	GREEN	BLUE	BLUE	BLUE	BLUE
	YELLOW ARROW	ORANGE (WH TT)	ORANGE (WH TT)	ORANGE (WH TT)	ORANGE (WH TT)
	GREEN ARROW	BLUE (WH TT)	BLUE (WH TT)	BLUE (WH TT)	BLUE (WH TT)
C	RED	RED	RED	RED (BL TT)	RED (BL TT)
	YELLOW	ORANGE	ORANGE (BL TT)	ORANGE	ORANGE (BL TT)
	GREEN	BLUE	BLUE	BLUE	BLUE
	YELLOW ARROW	ORANGE (WH TT)	ORANGE (WH TT)	ORANGE (WH TT)	ORANGE (WH TT)
	GREEN ARROW	BLUE (WH TT)	BLUE (WH TT)	BLUE (WH TT)	BLUE (WH TT)
D	RED	RED			
	YELLOW	ORANGE (RD TT)			
	GREEN	BLUE			
	YELLOW ARROW	ORANGE (WH TT)			
	GREEN ARROW	BLUE (WH TT)			
PA	DON'T WALK	YELLOW	YELLOW (RD TT)	YELLOW (OR TT)	YELLOW (BL TT)
	WALK	BLUE	BLUE	BLUE	BLUE
	PEDESTRIAN PUSHBUTTON	PURPLE (Y TT)	PURPLE (RD TT)	PURPLE (OR TT)	PURPLE (BL TT)
		PURPLE (Y TT)	PURPLE (RD TT)	PURPLE (OR TT)	PURPLE (BL TT)
PB	DON'T WALK	BROWN	BROWN (OR TT)	BROWN (RD TT)	BROWN (BL TT)
	WALK	BLUE	BLUE	BLUE	BLUE
	PEDESTRIAN PUSHBUTTON	PURPLE (BR TT)	PURPLE (OR TT)	PURPLE (RD TT)	PURPLE (BL TT)
		PURPLE (BR TT)	PURPLE (OR TT)	PURPLE (RD TT)	PURPLE (BL TT)
PC	DON'T WALK	ORANGE	ORANGE (BL TT)		
	WALK	BLUE	BLUE		
	PEDESTRIAN PUSHBUTTON	PURPLE (OR TT)	PURPLE (BL TT)		
		PURPLE (OR TT)	PURPLE (BL TT)		
PD	DON'T WALK	RED (BL TT)			
	WALK	BLUE			
	PEDESTRIAN PUSHBUTTON	PURPLE			
		PURPLE			
FS	RED	RED			
	YELLOW	YELLOW			

'FS' INDICATES 'FIRE SIGNAL' HEAD

TT – DENOTES TRACER TAPE
WH – WHITE
BL – BLUE
OR – ORANGE
RD – RED
BR – BROWN

NOTES:

1. SEE DRAWING SP635–2.5.7 FOR NOTES AND ADDITIONAL COLOUR CODING.

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C		
B	GENERAL REVISIONS	APR 17
A	GENERAL REVISIONS	AUG 94

SINGLE CONDUCTOR COLOUR CODING

Date Approved
30/09/93 E.L. (Signature on File)
Chief Highway Engineer

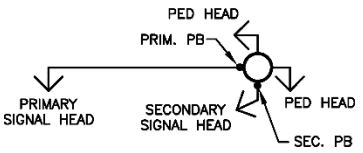
SPECIFICATION DRAWING No.
SP635–2.5.8

COLOUR CODING (MULTICONDUCTOR SIGNAL CABLE)

19, 21 OR 25 CONDUCTOR No. 14 (41 STRAND) CSA SPEC. C22.2 No. 210

CONDUCTOR No.	SIGNAL ASSIGNMENT	LETTERING	CONDUCTOR COLOUR	ALTERNATE COLOUR CODE	SINGLE CONDUCTOR COLOUR IN POLE
1	NEUTRAL	WHITE ONE	WHITE	WHITE	WHITE
2	PRIMARY PB RETURN	WHITE TWO	WHITE	BLACK, RED T.T.	PURPLE
3	PRIMARY PB	–	BLACK	BLACK, WHITE T.T.	PURPLE
4	SECONDARY PB	–	ORANGE	BLACK, ORANGE T.T.	PURPLE
5	PRIMARY RED	RED ONE	RED	RED	RED*
6	SECONDARY RED	RED TWO	RED	RED (BLACK T.T.)	RED*
7	SECONDARY PB RETURN	RED THREE	RED		PURPLE
8	PRIMARY PED DW	RED FOUR	RED	YELLOW, RED T.T.	YELLOW, BROWN, ORANGE OR RED(*)
9	SECONDARY PED DW	RED FIVE	RED	BROWN (ORANGE T.T.)	YELLOW, BROWN, ORANGE OR RED(*)
10	PRIMARY YELLOW	YELLOW ONE	YELLOW	YELLOW	YELLOW, BROWN OR ORANGE(*)
11	SECONDARY YELLOW	YELLOW TWO	YELLOW	BROWN (BLACK T.T.)	YELLOW, BROWN OR ORANGE(*)
12	PRIMARY LT YELLOW	YELLOW THREE	YELLOW	ORANGE, WHITE T.T.	ORANGE(*)
13	SECONDARY LT YELLOW	YELLOW FOUR	YELLOW	ORANGE, WHITE T.T.	ORANGE(*)
14	SECONDARY PED WALK	YELLOW FIVE	YELLOW	BLUE, ORANGE T.T.	BLUE(*)
15	PRIMARY GREEN	GREEN ONE	BLUE	BLUE	BLUE(*)
16	SECONDARY GREEN	GREEN TWO	BLUE	BLUE (BLACK T.T.)	BLUE(*)
17	PRIMARY LT GREEN	GREEN THREE	BLUE	BLUE (WHITE T.T.)	BLUE(*)
18	SECONDARY LT GREEN	GREEN FOUR	BLUE		BLUE(*)
19	PRIMARY PED WALK	GREEN FIVE	BLUE	BLUE, RED T.T.	BLUE(*)
20	PRIMARY LT RED	RED SIX	RED		RED (BLUE T.T.)
21	SECONDARY LT RED	RED SEVEN	RED		RED (BLUE T.T.)
22	SPARE/PRE-EMPT	AMBER SIX	YELLOW	YELLOW – GREEN T.T	–
23	SPARE/PRE-EMPT	AMBER SEVEN	YELLOW	BLUE – GREEN T.T.	–
24	SPARE	GREEN SIX	BLUE	ORANGE	–
25	SPARE	GREEN SEVEN	BLUE	ORANGE/BLACK TRACE	–
	GROUND			GREEN	

LT = LEFT TURN SIGNAL
DW = DON'T WALK
PB = PEDESTRIAN PUSHBUTTON
TT = TRACER TAPE
* YELLOW or BROWN DESIGNATIONS – YELLOW (N/B & S/B) BROWN – (E/B & W/B)



(*) SEE NOTE 3

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. COLOUR CODING AND TRACER TAPE INSIDE POLES SHALL BE IN ACCORDANCE WITH SP635-2.5.7 & 2.5.8.

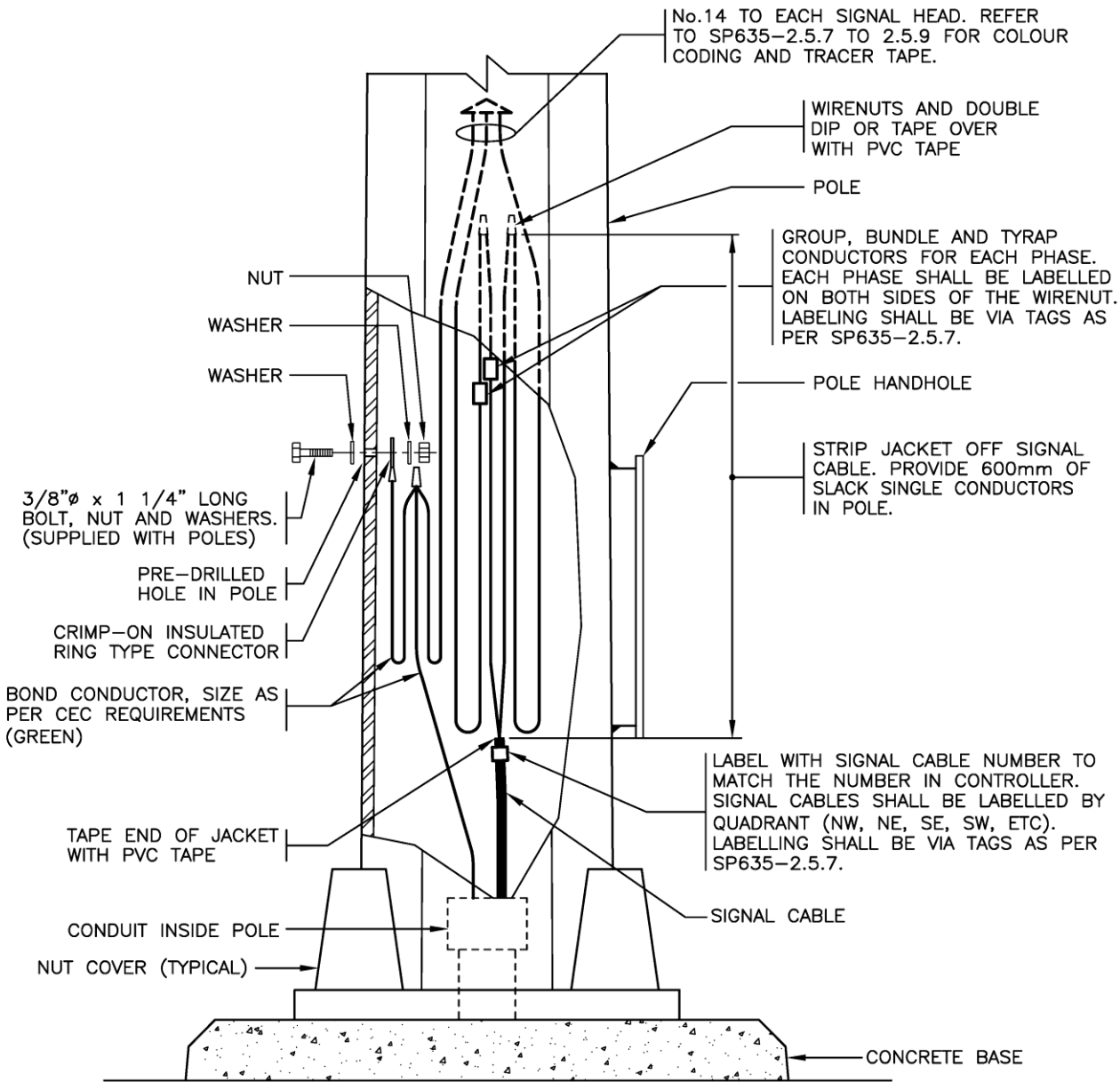
BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
E		
D		
C		
B	CSA SPEC UPDATE & 21C CABLE ADDED NOTE 2 REMOVED	APR 20
A	GENERAL REVISIONS	APR 17

SIGNAL CABLE WIRING & COLOUR CODING

Date: **14/11/03** Approved: **D.N. (Signature on File)** Chief Engineer

SPECIFICATION DRAWING No. SP635-2.5.9



ELEVATION

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. REFER TO SP635–2.5.7 FOR CONDUCTOR LABELING.

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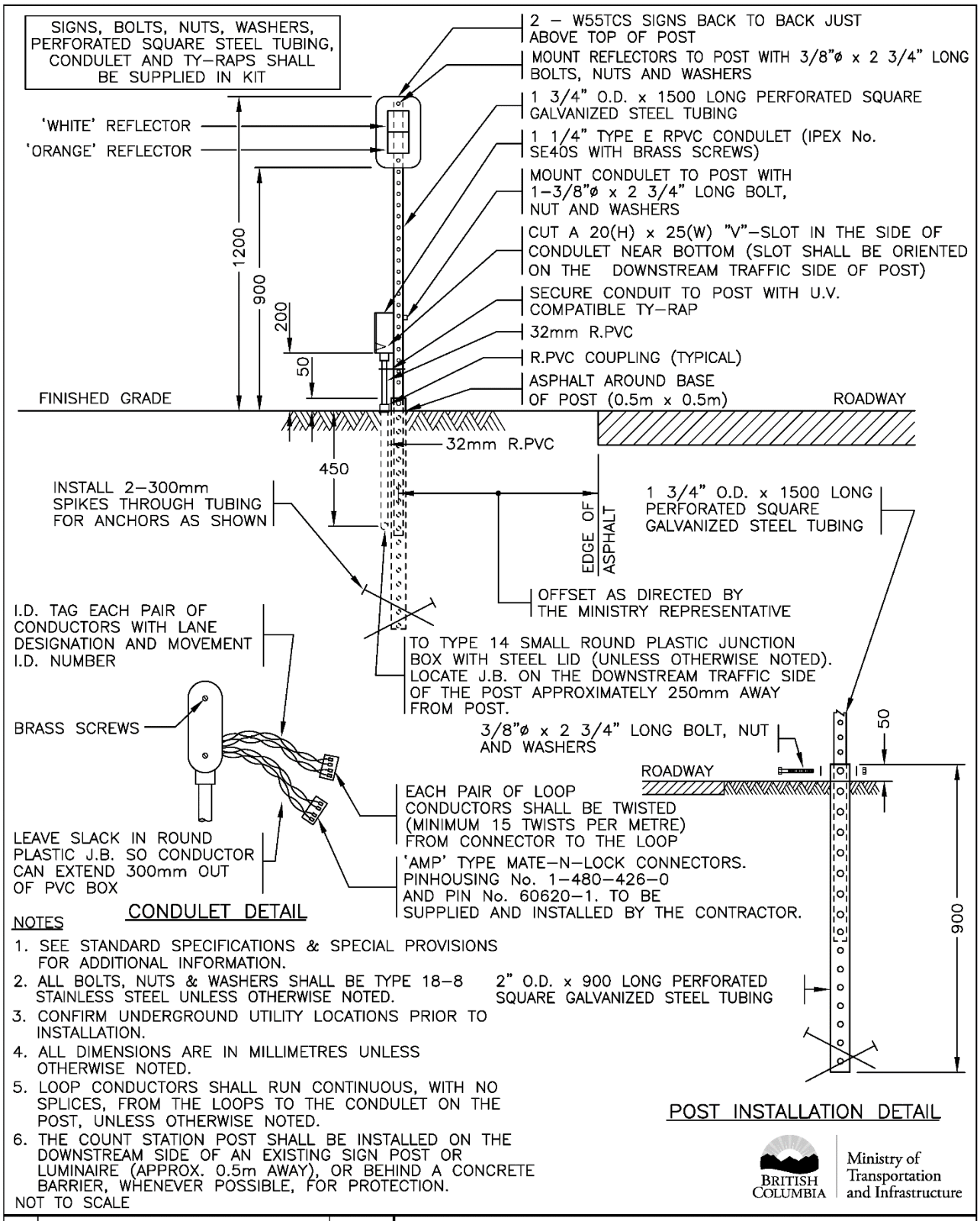
NOT TO SCALE

No.	Revision	Date
F		
E		
D		
C	BOND CONDUCTOR REVISED	JUN 22
B	GENERAL REVISIONS	APR 20
A	GENERAL REVISIONS	APR 17

SIGNAL CABLE WIRING IN POLE HANDHOLE

Date: **14/11/03** Approved: **D.N.** (Signature on File) — Chief Engineer

SPECIFICATION DRAWING No. SP635–2.5.10



CONDULET DETAIL

POST INSTALLATION DETAIL

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL BOLTS, NUTS & WASHERS SHALL BE TYPE 18–8 STAINLESS STEEL UNLESS OTHERWISE NOTED.
3. CONFIRM UNDERGROUND UTILITY LOCATIONS PRIOR TO INSTALLATION.
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
5. LOOP CONDUCTORS SHALL RUN CONTINUOUS, WITH NO SPLICES, FROM THE LOOPS TO THE CONDULET ON THE POST, UNLESS OTHERWISE NOTED.
6. THE COUNT STATION POST SHALL BE INSTALLED ON THE DOWNSTREAM SIDE OF AN EXISTING SIGN POST OR LUMINAIRE (APPROX. 0.5m AWAY), OR BEHIND A CONCRETE BARRIER, WHENEVER POSSIBLE, FOR PROTECTION.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
I	GENERAL REVISIONS	APR 20
H	GENERAL REVISIONS	APR 17
G	COLD PATCH ADDED & UPPER SPIKES REMOVED	NOV 01
F	GENERAL REVISIONS	SEPT 00
E	NOTE 6 ADDED	AUG 99
D	KIT NOTE ADDED, NOTES REVISED	NOV 98

SHORT COUNT STATION INSTALLATION DETAILS

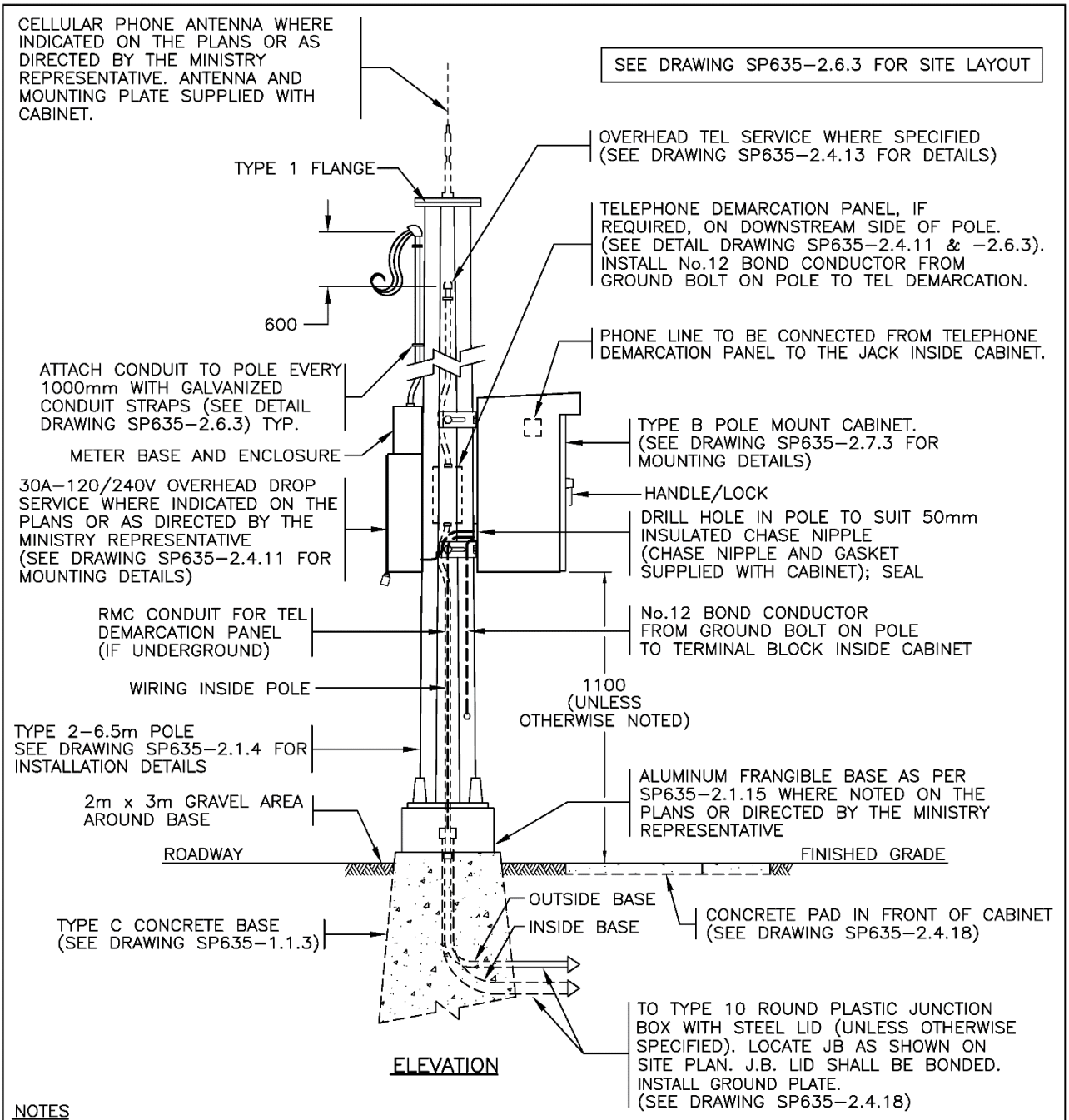
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.6.1



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. CONTRACTOR TO CONFIRM ORIENTATION OF CABINET WITH THE MINISTRY REPRESENTATIVE PRIOR TO INSTALLATION.
3. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
4. 2m OF LOOP WIRE AND/OR SHIELDED CABLE SHALL BE COILED IN THE CABINET FOR CONNECTIONS BY MINISTRY. TAG AND LABEL CABLES IN ACCORDANCE WITH DRAWING SP635-2.5.7. THE CABINET LOOP OPENING SHALL BE DUCT-SEALED ONCE WIRING IS PULLED THRU.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F	GENERAL REVISIONS	APR 20
E	GENERAL REVISIONS	APR 17
D	GENERAL REVISIONS	SEPT 00
C	TEL DEMARCATION PANEL RELOCATED	AUG 99
B	TEL DEMARCATION & CONCRETE PADS SHOWN	NOV 98
A	O/H TEL & POWER SERVICES ADDED	OCT 97

PERMANENT COUNT STATION INSTALLATION DETAILS

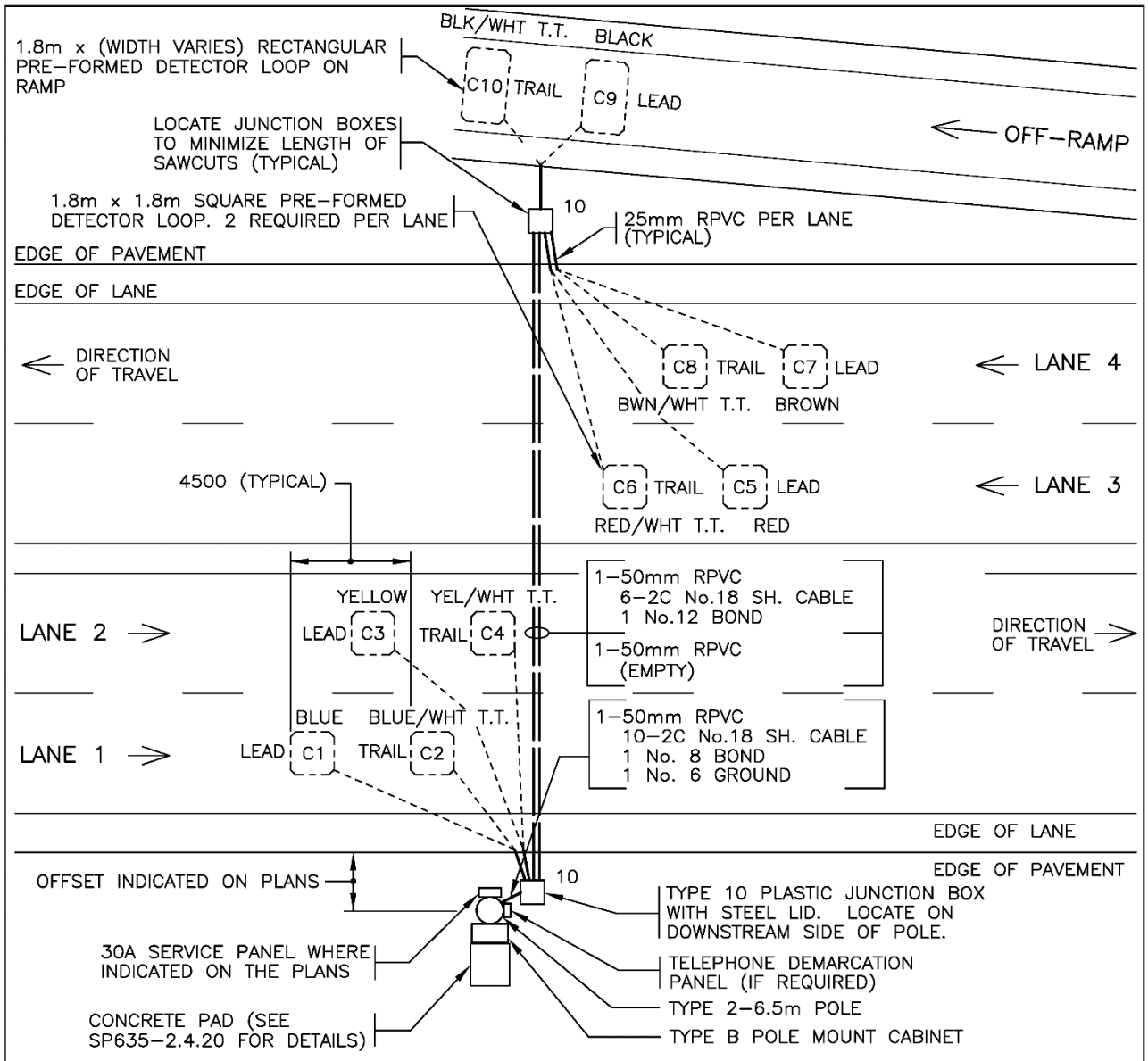
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-2.6.2



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. CONTRACTOR TO CONFIRM ORIENTATION OF CABINET AND SITE PLAN LAYOUT WITH THE MINISTRY REPRESENTATIVE PRIOR TO INSTALLATION.
3. 1-25mm RPVC LOOP CONDUIT REQUIRED PER LANE.
4. LOOPS SHALL BE NUMBERED SEQUENTIALLY FROM CABINET SIDE OUT.
5. LOOPS SHALL BE CUT 4500mm FROM LEADING EDGE OF THE LEADING LOOP TO THE LEADING EDGE OF THE TRAILING LOOP PER LANE.
6. LOCATE LOOPS IN CENTRE OF LANES UNLESS OTHERWISE NOTED ON THE PLANS.

SEE DRAWING SP635-2.6.2 FOR POLE ELEVATION

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F	GENERAL REVISIONS	APR 20
E	GENERAL REVISIONS	APR 17
D	GRAVEL CHANGED TO CRUSH IN NOTE 9	NOV 01
C	CONDUIT CROSSING & NOTES REVISED	SEPT 00
B	LOOP LAYOUT REVISED	AUG 99
A	LOOP LAYOUT REVISED	NOV 98

SITE PLAN AND LOOP LAYOUT FOR PERMANENT COUNT STATIONS

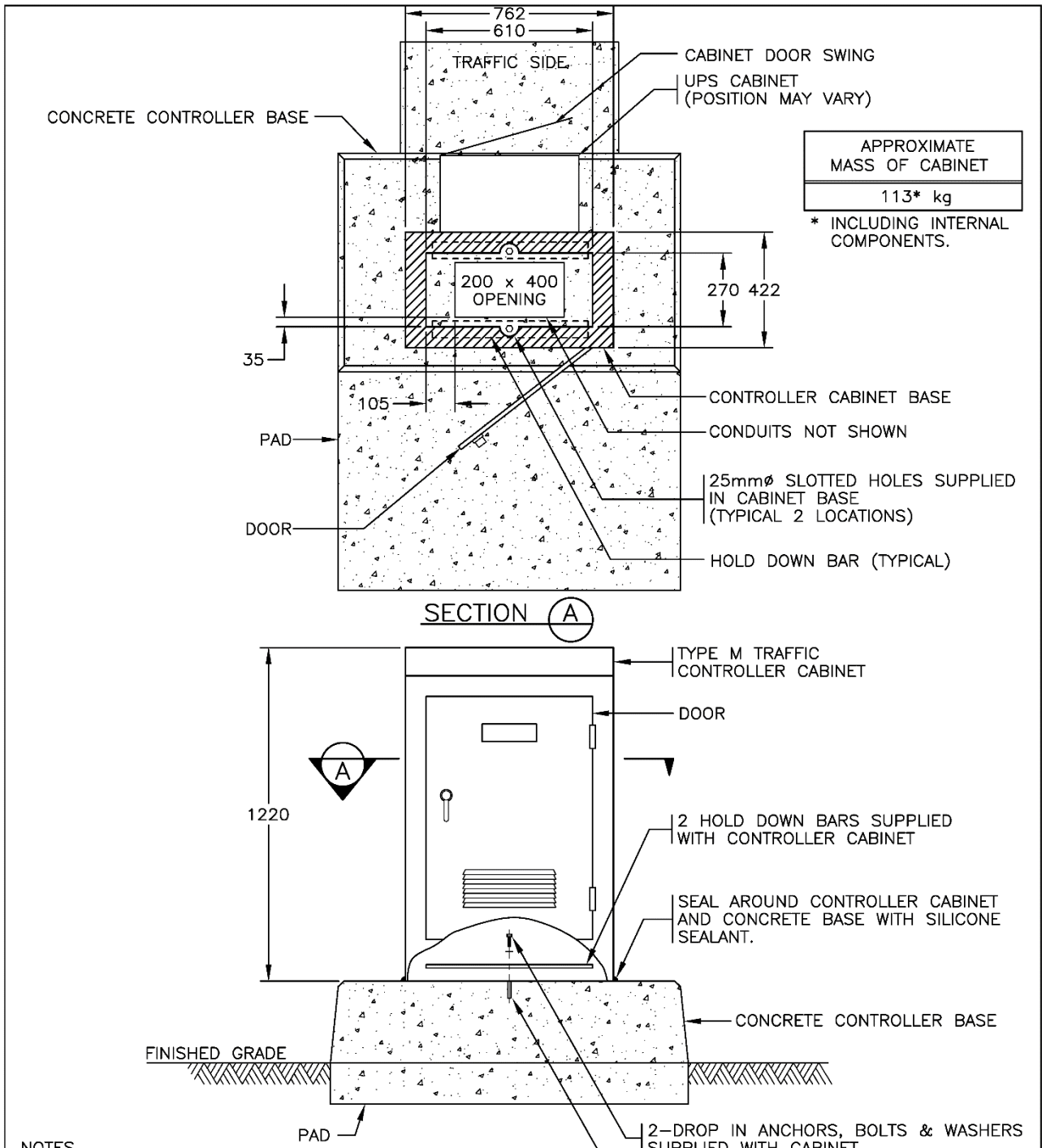
Date Approved

18/02/98 D.M. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-2.6.3



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. SEE DRAWING SP635–1.1.38 FOR BASE DIMENSIONS.
3. SEE DRAWING SP635–1.1.40 FOR BASE INSTALLATION DETAILS.
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.

NOT TO SCALE

BRITISH COLUMBIA — Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C	GENERAL REVISIONS	APR 20
B	GENERAL REVISIONS	APR 17
A	GENERAL REVISIONS	AUG 94

TYPE M CONTROLLER CABINET INSTALLATION DETAILS

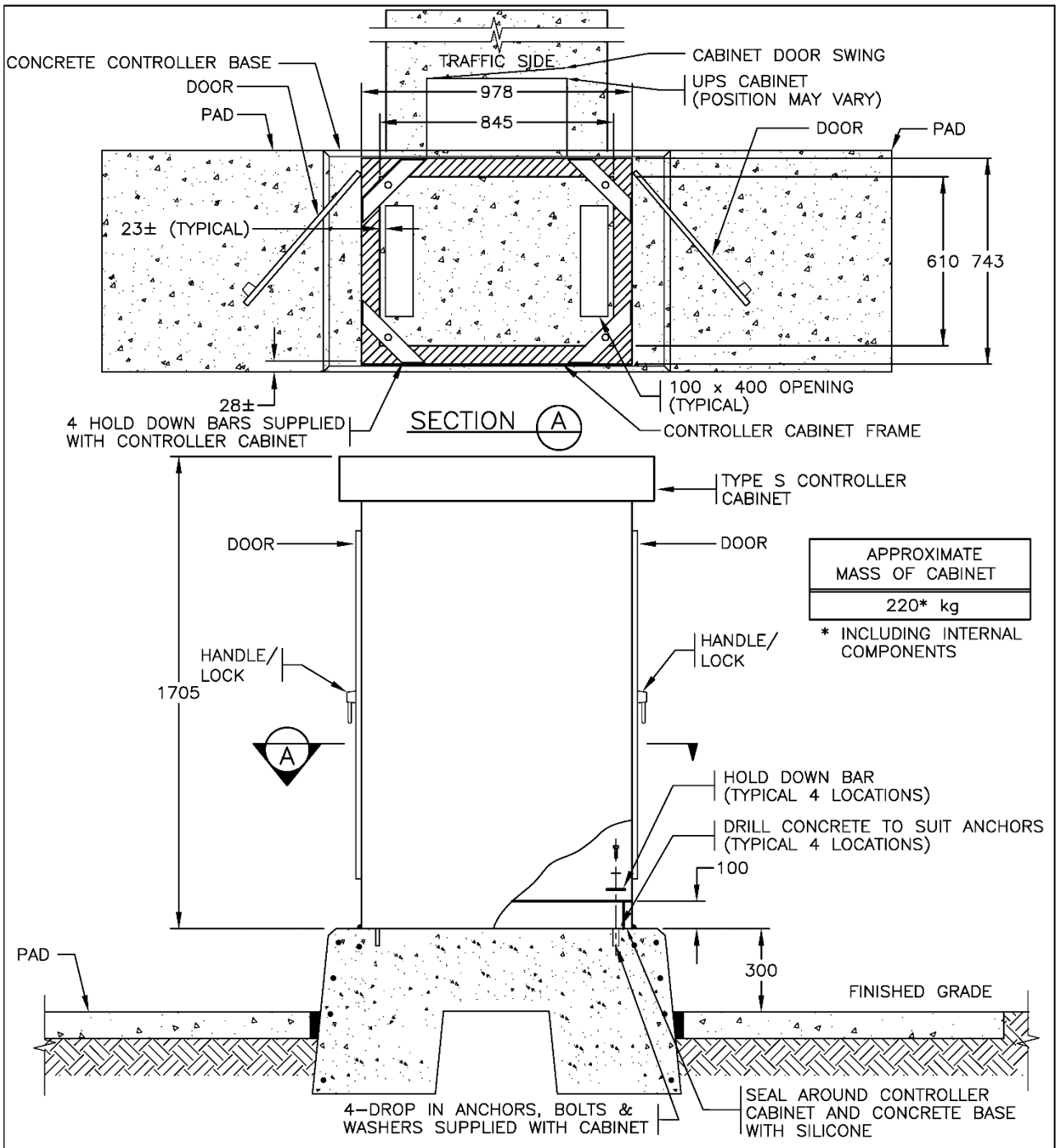
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.7.1



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. SEE DRAWING SP635–1.1.38 FOR BASE DIMENSIONS.
3. SEE DRAWING SP635–1.1.41 FOR BASE INSTALLATION DETAILS.
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C	GENERAL REVISIONS	APR 20
B	GENERAL REVISIONS	APR 17
A	GENERAL REVISIONS	AUG 94

TYPE S CONTROLLER CABINET INSTALLATION DETAILS

Date Approved

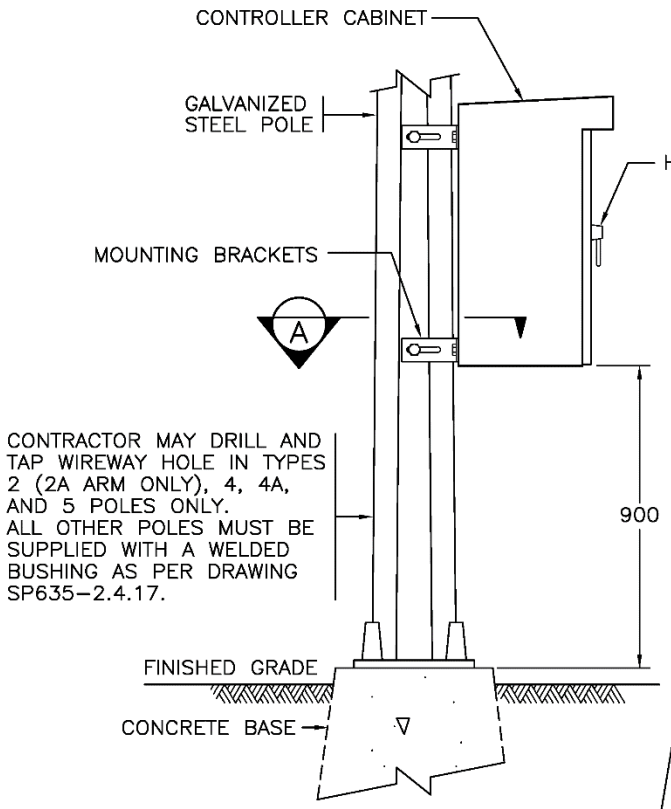
30/09/93 E.L. **(Signature on File)**

Chief Highway Engineer

SPECIFICATION DRAWING No.

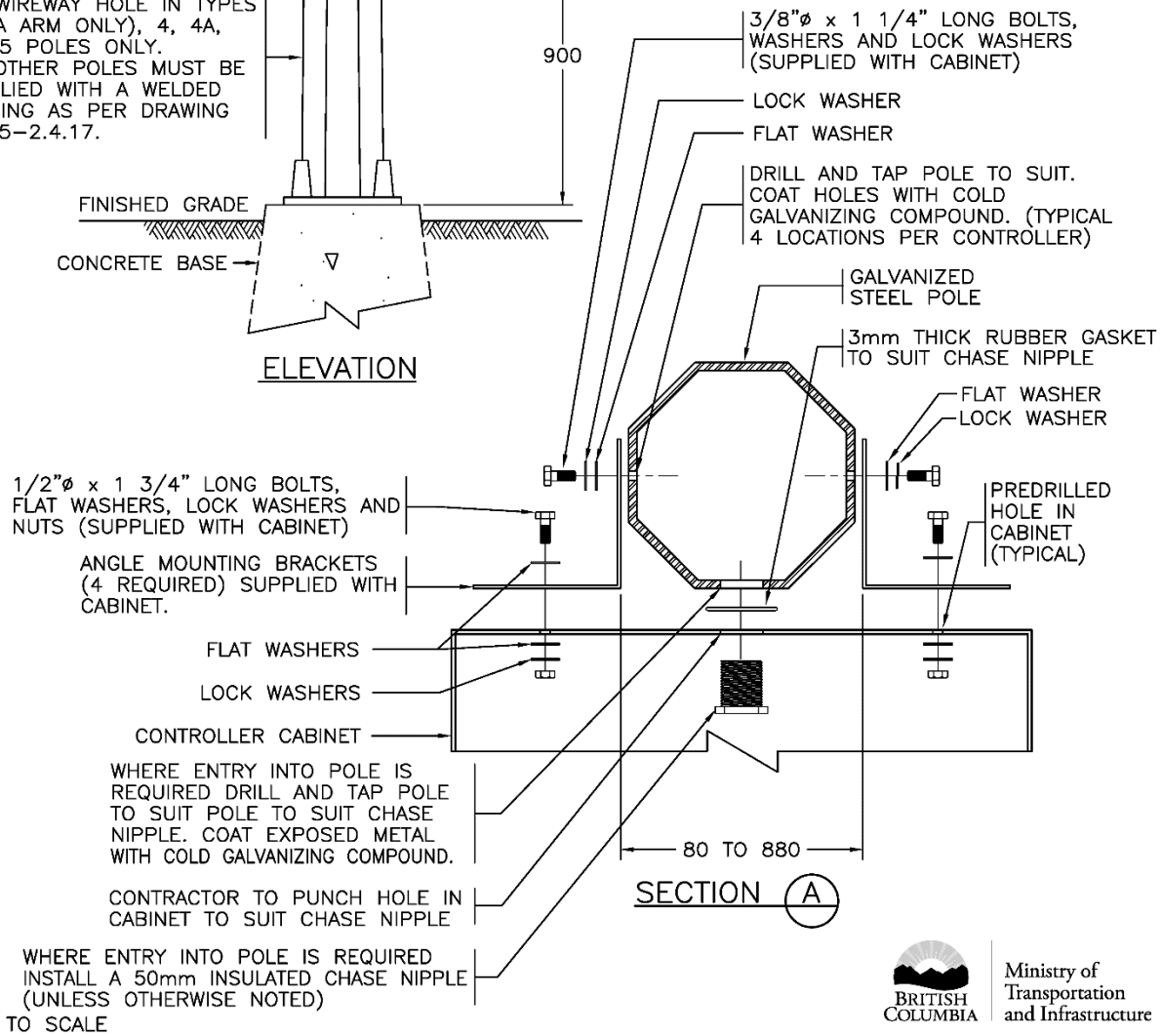
SP635–2.7.2

CABINET TYPE	HEIGHT	WIDTH	DEPTH	APPROXIMATE MASS
B	883	556	406	37 kg
C	1040	556	406	56 kg



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ORIENTATION OF CABINET SHALL BE CONFIRMED BY THE MINISTRY ELECTRICAL REPRESENTATIVE PRIOR TO INSTALLATION.
3. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.



NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	GENERAL REVISIONS	APR 17
C	TYPE G CABINET DELETED	OCT 03
B	MOUNTING HEIGHT REVISED	OCT 97
A	GENERAL REVISIONS	AUG 94

TYPES B AND C CONTROLLER CABINET INSTALLATION DETAILS

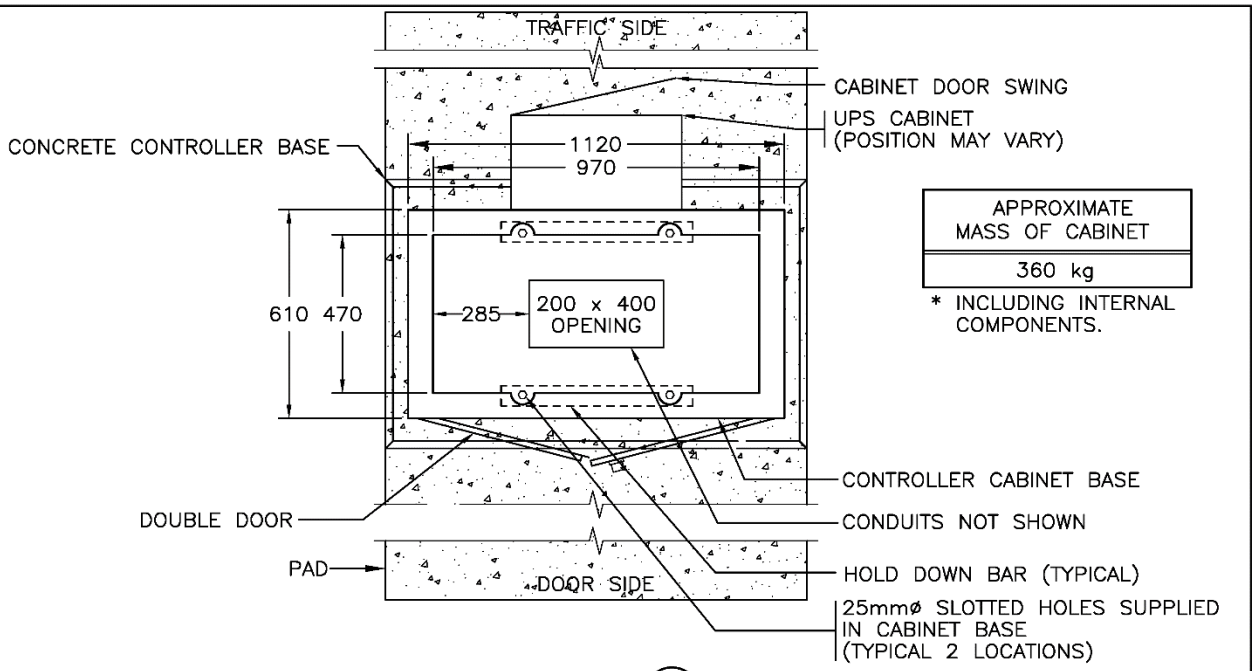
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

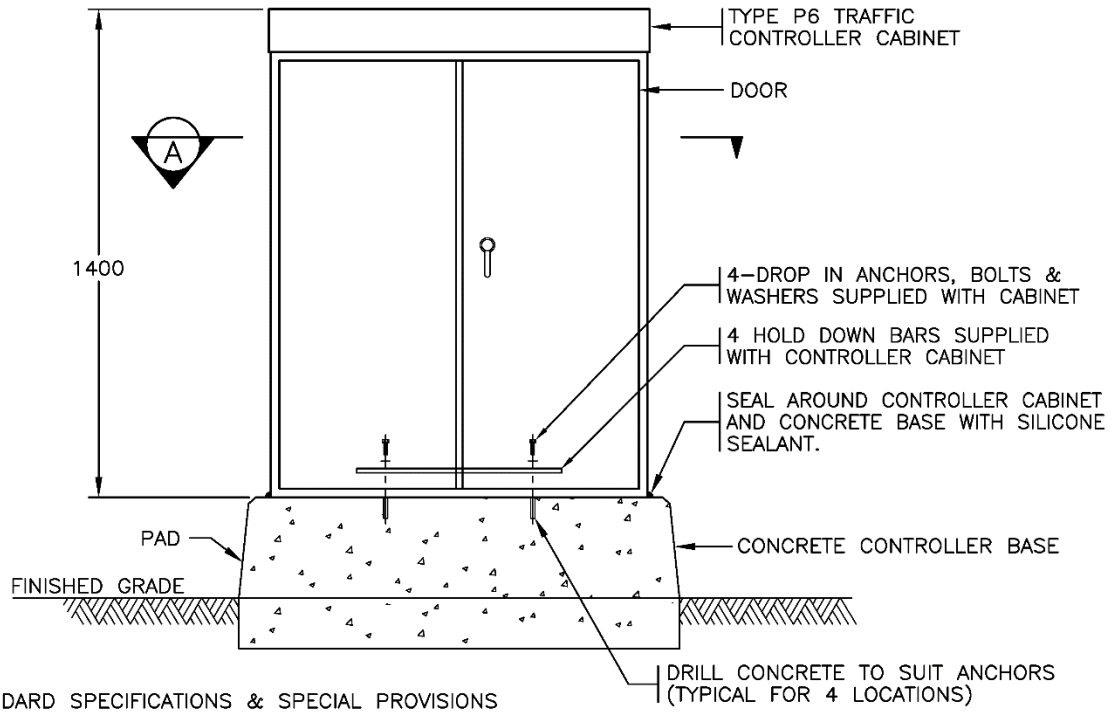
SP635-2.7.3



APPROXIMATE MASS OF CABINET
360 kg

* INCLUDING INTERNAL COMPONENTS.

SECTION A



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. SEE DRAWING SP635–1.1.38 FOR BASE DIMENSIONS.
3. SEE DRAWING SP635–1.1.42 FOR BASE INSTALLATION DETAILS.
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C		
B		
A	GENERAL REVISIONS	APR 20

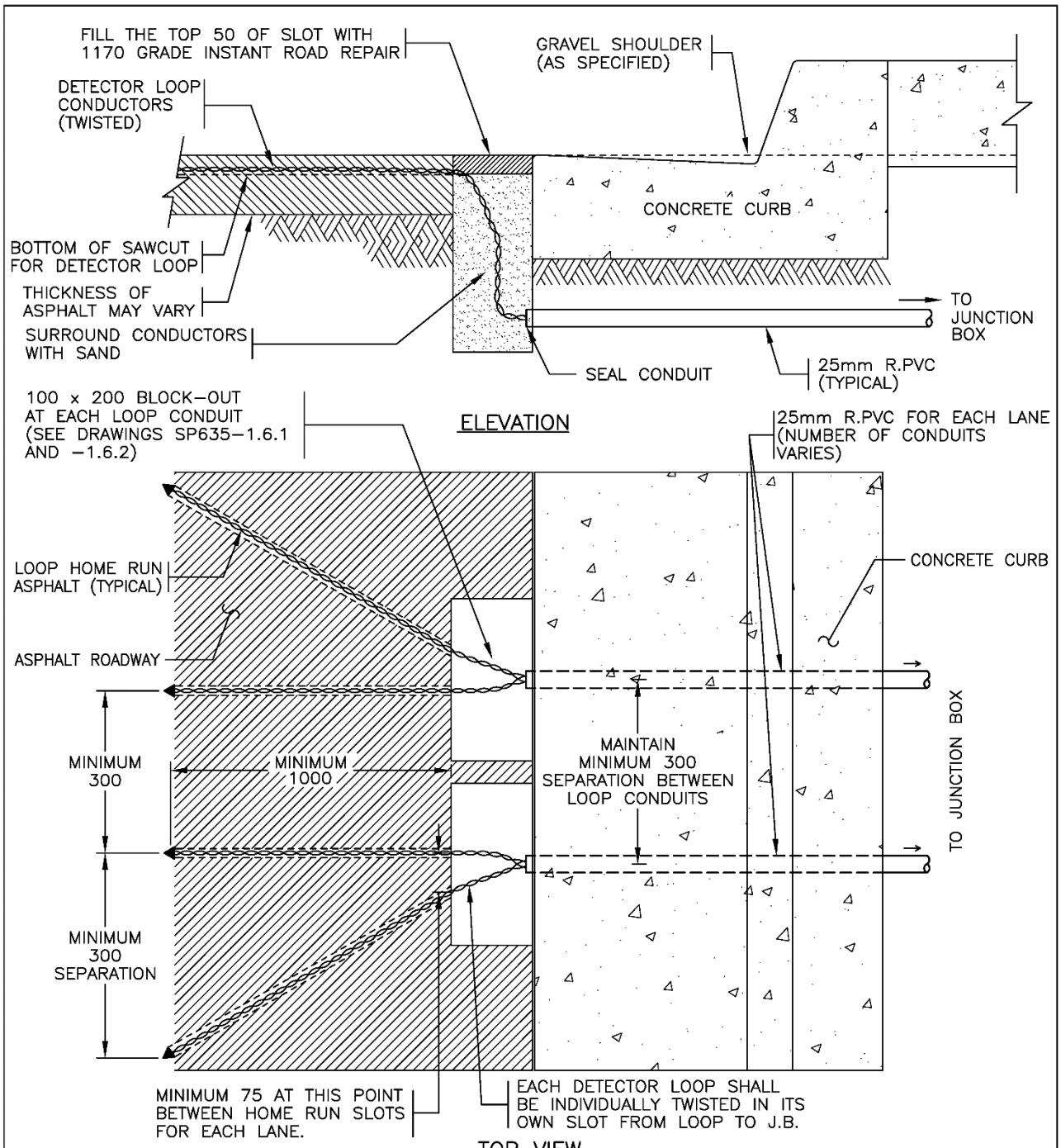
TYPE P6 CONTROLLER CABINET INSTALLATION DETAILS

Date Approved

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.7.4



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. SEE DRAWING SP635-2.8.6 FOR LOOP COLOR CODING.
3. SEE DRAWING SP635-1.6.1 & 1.6.2 FOR LOOP CONDUIT INSTALLATION DETAILS.
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.

NOT TO SCALE

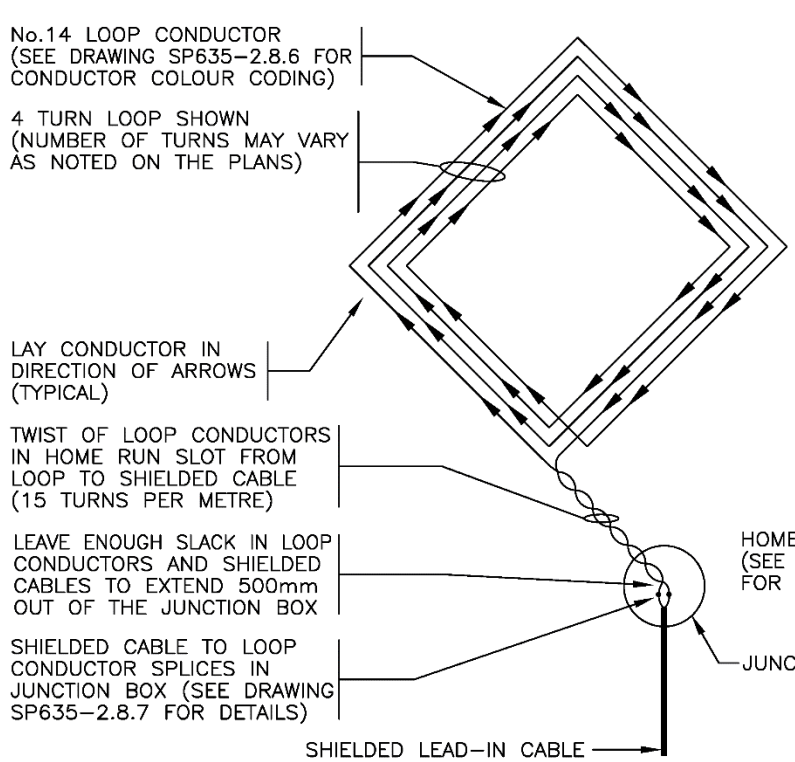
BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	GENERAL REVISIONS	APR 20
C	GENERAL REVISIONS	APR 17
B	REFERENCED DWG. No. REV'D (NOTE 2)	DEC 98
A	SPELLING REVISION	AUG 96

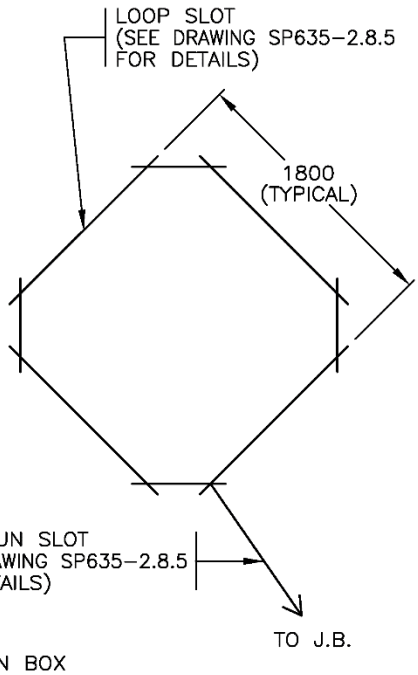
DETECTOR LOOP INSTALLATION DETAILS

Date: **15/11/95** Approved: **M.C.** (Signature on File)
Chief Highway Engineer

SPECIFICATION DRAWING No.
SP635-2.8.1



LOOP CONDUCTOR LAYOUT



LOOP SLOT PATTERN

LOOP INDUCTANCE TABLE (μH) (LESS SHIELDED LEAD-IN CABLE INDUCTANCE)	
TURNS	INDUCTANCE
4	120 μH
5	180 μH
6	252 μH
7	336 μH

LOOP INDUCTANCE SHALL BE GREATER THAN THE SHIELDED CABLE INDUCTANCE. (A 2:1 RATIO IS PREFERABLE). SHIELDED CABLE INDUCTANCE IS 0.56 μH PER METRE FOR 18 AWG WIRE.

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. LOOP CONDUCTORS SHALL BE INSTALLED IN THE SAWCUT SLOT.
4. LOOP INSTALLATION PROCEDURES & RULES SHALL BE FOLLOWED IN ACCORDANCE WITH DRAWINGS SP635-2.8.8 AND -2.8.9.
5. ALL DIAMOND DETECTOR LOOPS SHALL BE A MINIMUM OF 4 TURNS.
6. DIAMOND LOOPS SHALL NOT BE USED FOR SHORT DURATION OR PERMANENT TRAFFIC COUNTER STATION INSTALLATIONS.

NOT TO SCALE

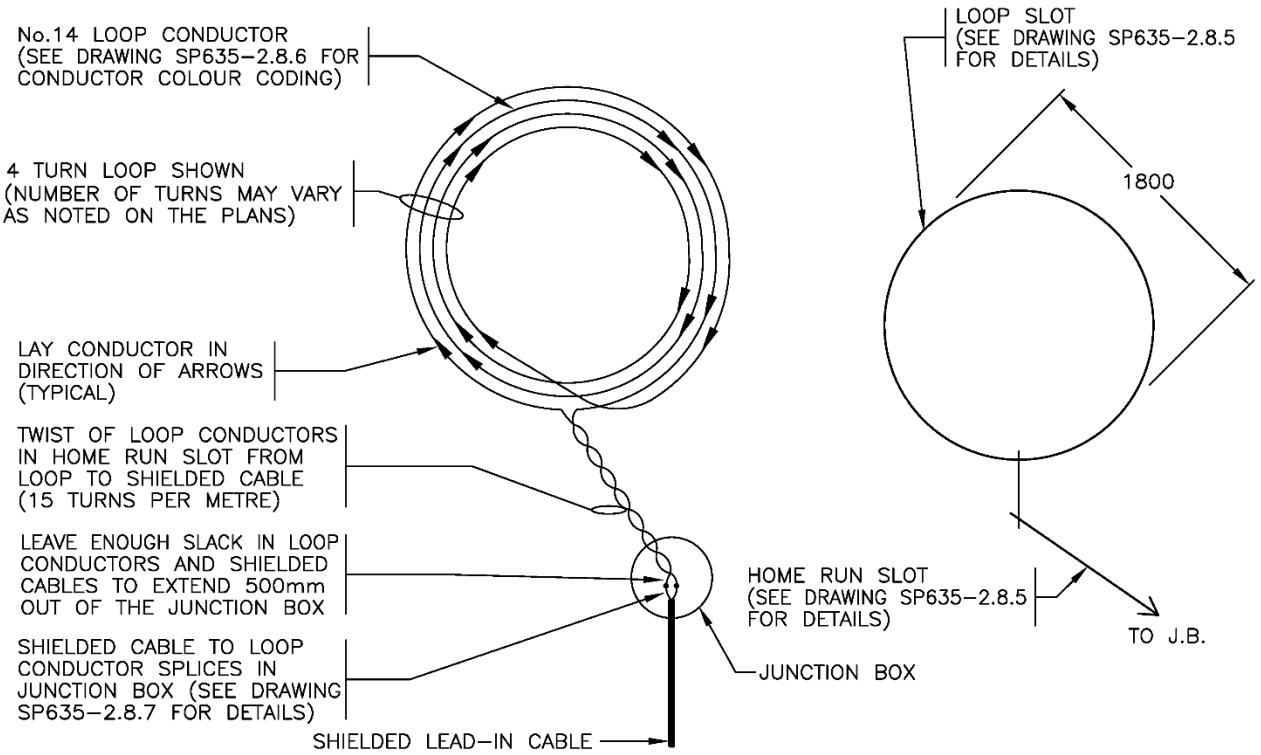
BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	GENERAL REVISIONS	APR 20
C	GENERAL REVISIONS	APR 17
B	SHIELDED CABLE WIRE SIZE NOTED, NOTE 6 ADDED	NOV 98
A	SHIELDED CABLE INDUCTANCE REVISED	AUG 96

DIAMOND DETECTOR LOOP INSTALLATION DETAILS

Date: **15/11/95** Approved: **M.C.** **(Signature on File)**
Chief Highway Engineer

SPECIFICATION DRAWING No.
SP635-2.8.2



LOOP CONDUCTOR LAYOUT

LOOP SLOT PATTERN

LOOP INDUCTANCE TABLE (μH) (LESS SHIELDED LEAD–IN CABLE INDUCTANCE)	
TURNS	INDUCTANCE
4	95 μH
5	140 μH
6	195 μH
7	260 μH

LOOP INDUCTANCE SHALL BE GREATER THAN THE SHIELDED CABLE INDUCTANCE. (A 2:1 RATIO IS PREFERABLE). SHIELDED CABLE INDUCTANCE IS 0.56 μH PER METRE FOR 18 AWG WIRE.

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. LOOP CONDUCTORS SHALL BE INSTALLED IN THE SAWCUT SLOT.
4. LOOP INSTALLATION PROCEDURES & RULES SHALL BE FOLLOWED IN ACCORDANCE WITH DRAWINGS SP635–2.8.8 AND –2.8.9.
5. ALL ROUND DETECTOR LOOPS SHALL BE A MINIMUM OF 4 TURNS.
6. ROUND LOOPS SHALL NOT BE USED FOR SHORT DURATION OR PERMANENT TRAFFIC COUNTER STATION INSTALLATIONS.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C	GENERAL REVISIONS	APR 20
B	GENERAL REVISIONS	APR 17
A	ALTERNATE NOTE REMOVED	OCT 03

ROUND DETECTOR LOOP INSTALLATION DETAILS

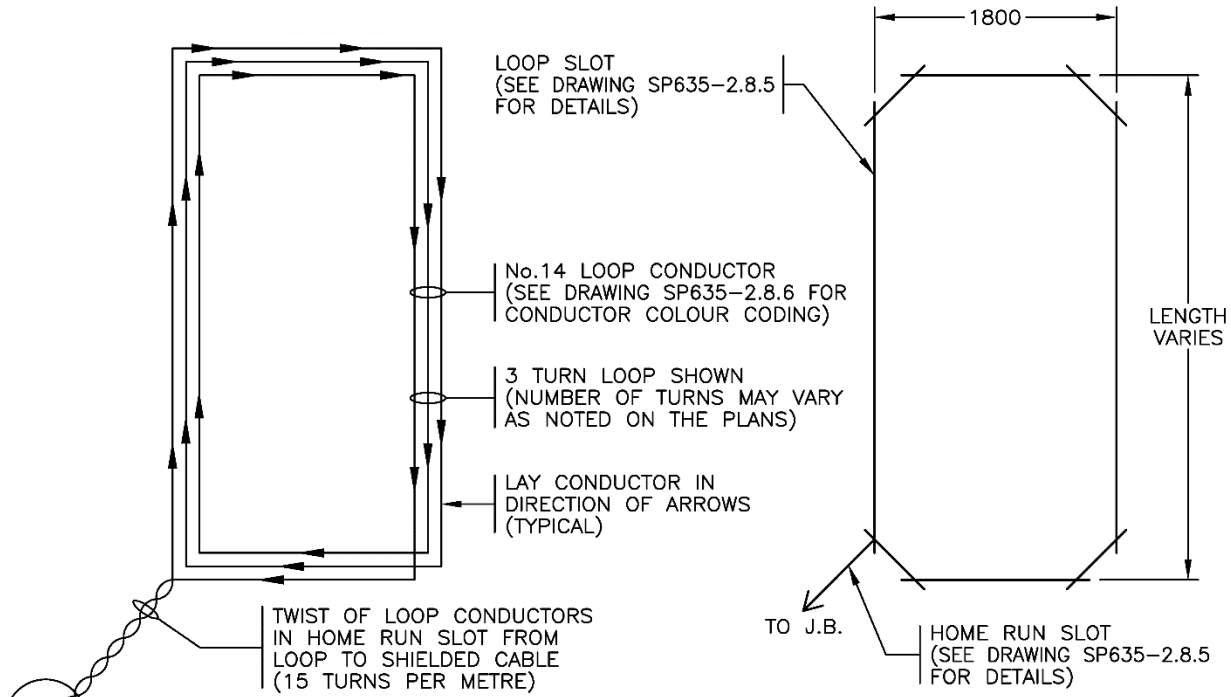
Date Approved

30/09/93 E.L. (Signature on File)

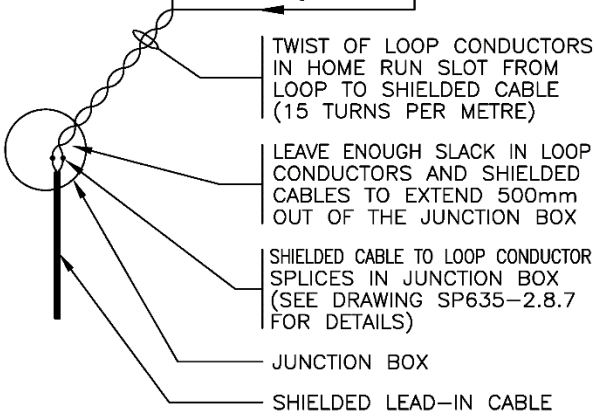
Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.8.3



LOOP SLOT PATTERN



LOOP CONDUCTOR LAYOUT

LOOP INDUCTANCE FORMULA (μH) (LESS SHIELDED LEAD-IN CABLE INDUCTANCE)
$L = P/4\ (3.28)(N^2+N)$ L = LOOP INDUCTANCE (μH) P = PERIMETER (METRES) N = NUMBER OF TURNS

LOOP INDUCTANCE SHALL BE GREATER THAN THE SHIELDED CABLE INDUCTANCE. (A 2:1 RATIO IS PREFERABLE). SHIELDED CABLE INDUCTANCE IS 0.56 μH PER METRE FOR 18 AWG WIRE.

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. LOOP CONDUCTORS SHALL BE INSTALLED IN THE SLOTTED SAWCUT.
4. LOOP INSTALLATION PROCEDURES & RULES SHALL BE FOLLOWED IN ACCORDANCE WITH DRAWINGS SP635-2.8.8 AND -2.8.9.
5. ALL RECTANGULAR DETECTOR LOOPS SHALL BE A MINIMUM OF 3 TURNS.

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

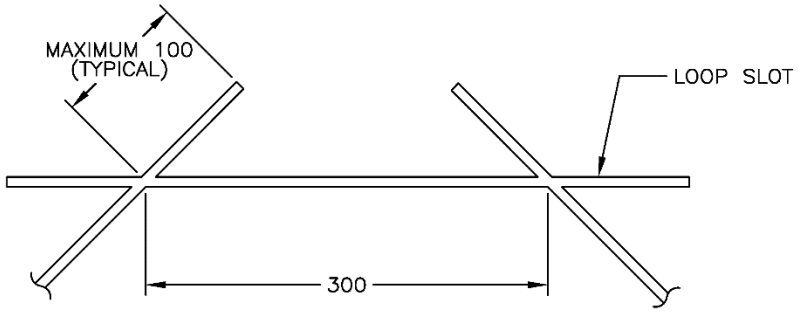
NOT TO SCALE

No.	Revision	Date
E		
D	GENERAL REVISIONS	APR 20
C	GENERAL REVISIONS	APR 17
B	DRAWING NUMBER CHANGED & SHIELDED CABLE WIRE SIZE NOTED	NOV 98
A	SHIELDED CABLE INDUCTANCE REVISED	AUG 96

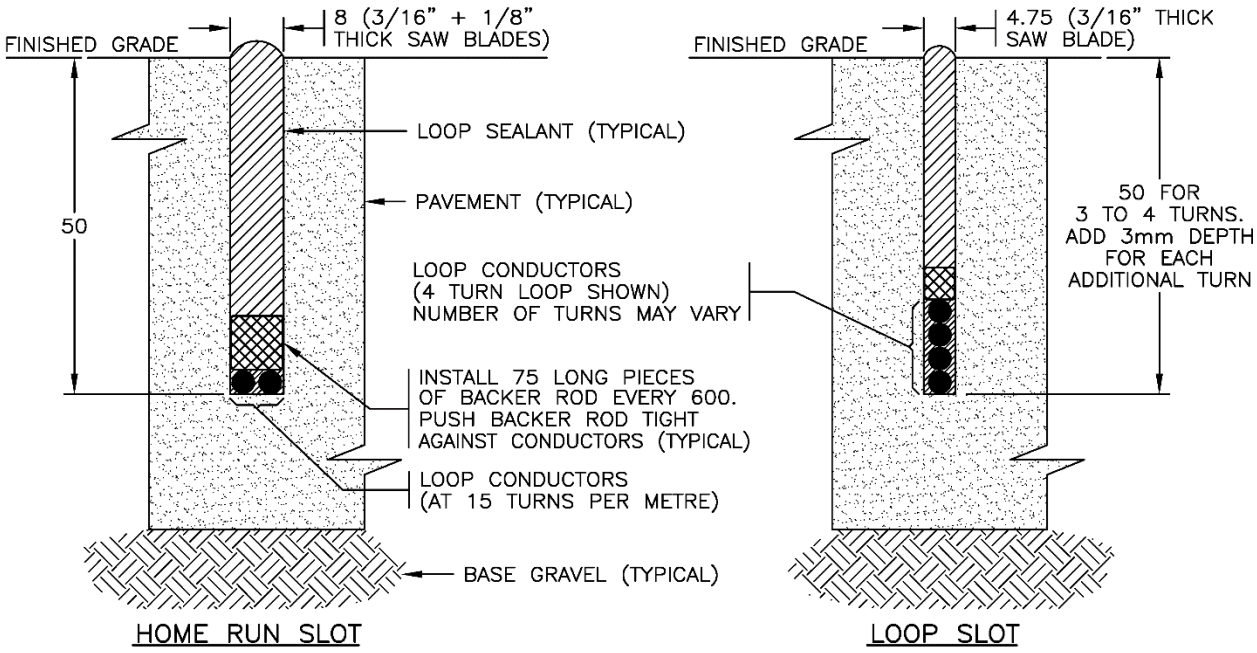
RECTANGULAR DETECTOR LOOP INSTALLATION DETAILS

Date: **15/11/95** Approved: **M.C.** **(Signature on File)**
Chief Highway Engineer

SPECIFICATION DRAWING No.
SP635-2.8.4



TYPICAL LOOP CORNER DETAIL
(FOR DIAMOND & RECTANGULAR LOOPS)



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES.
3. APPROVED LOOP SLOT SEALANTS MUST MEET ASTM D6690 TYPE 1. ALTERNATE PRODUCTS MUST MEET MINISTRY APPROVAL.
4. ONLY ONE LOOP SHALL BE INSTALLED IN EACH HOME RUN SLOT.

NOT TO SCALE

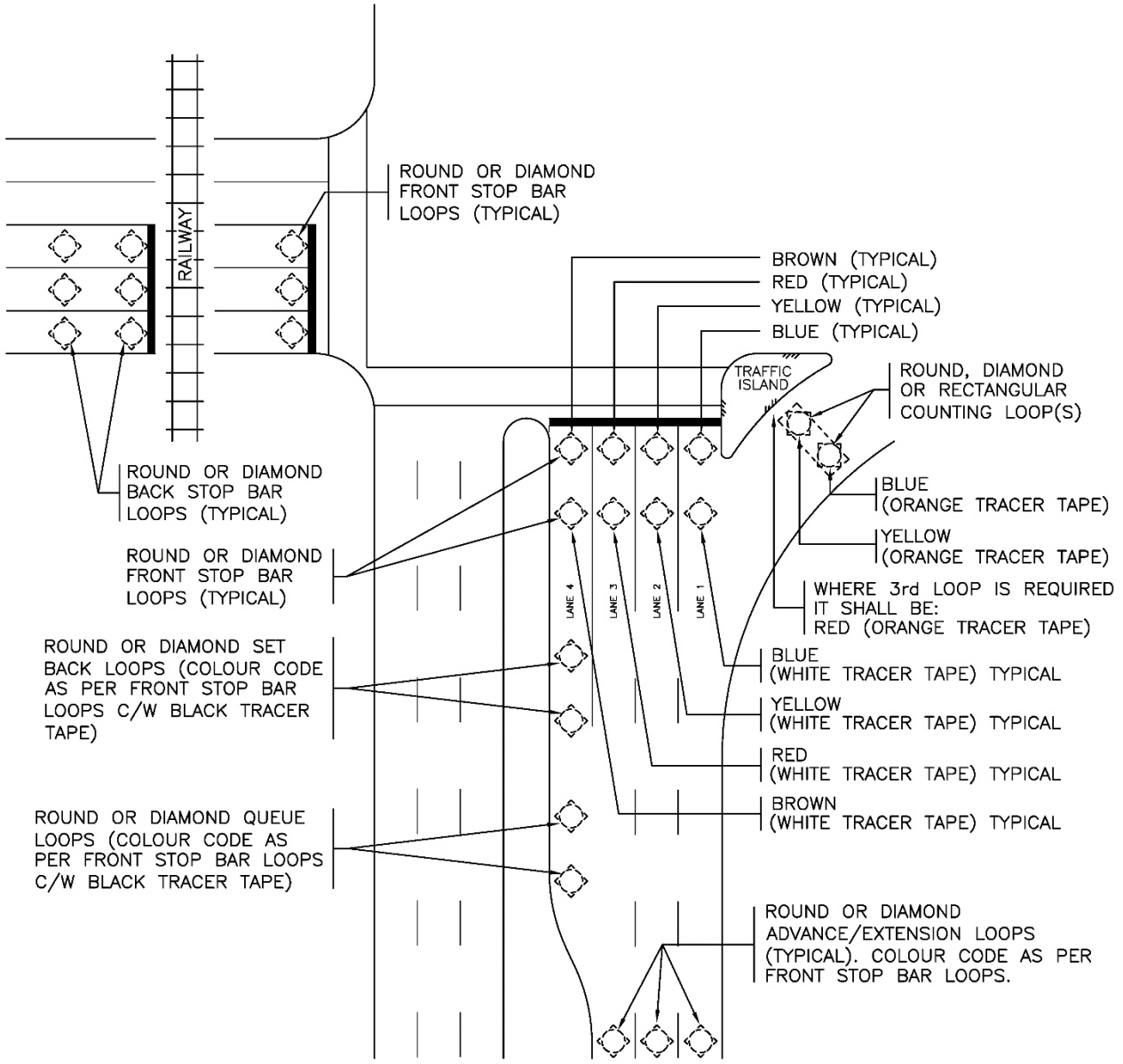
BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	NOTE 3 UPDATED. NOTE 4 REMOVED.	APR 20
C	GENERAL REVISIONS	APR 17
B	APPROVED SEALER ADDED	AUG 99
A	DRAWING NUMBER CHANGED	NOV 98

LOOP AND HOME RUN SLOT DETAILS

Date Approved
15/11/95 M.C. (Signature on File)
Chief Highway Engineer

SPECIFICATION DRAWING No.
SP635–2.8.5



DETECTOR LOOPS

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION
2. COLOUR CODE IS BASED ON USING RIGHT HAND CURB OF ROAD AS POINT OF REFERENCE & WORKING LOOP CONDUCTORS LEFT TOWARDS CENTRE OF ROADWAY. WHERE A 5th LANE, USE ORANGE CONDUCTOR COLOUR.
3. RECTANGULAR DETECTOR LOOP CONDUCTORS SHALL BE BLACK.

NOT TO SCALE

BRITISH COLUMBIA — Ministry of Transportation and Infrastructure

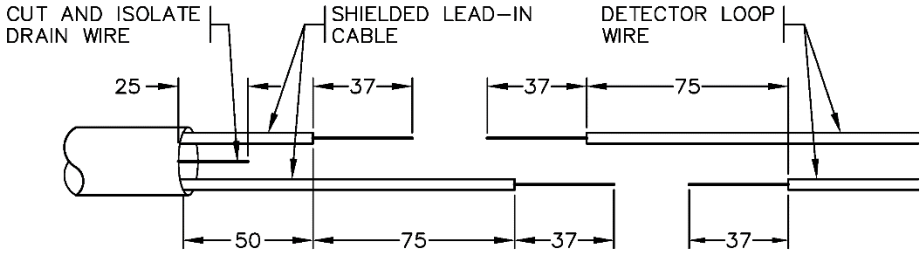
No.	Revision	Date
E		
D		
C	GENERAL REVISIONS	APR 17
B	SETBACK, QUEUE & ADVANCE LOOPS ADDED	SEPT 00
A	DRAWING NUMBER CHANGED, ROUND LOOPS ADDED	NOV 98

DETECTOR LOOP TYPES COLOUR CODING

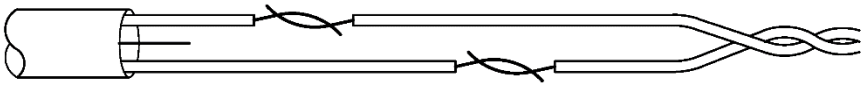
Date Approved
15/11/95 M.C. (Signature on File)
Chief Highway Engineer

SPECIFICATION DRAWING No.
SP635–2.8.6

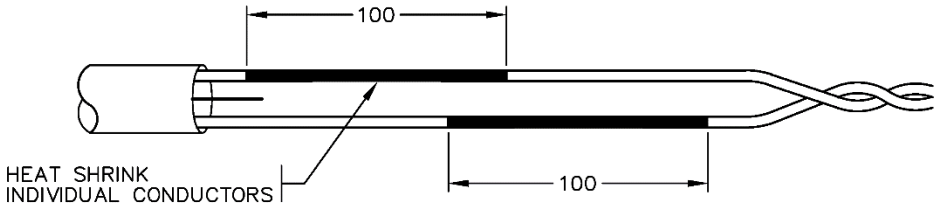
STEP 1. STRIP LOOP WIRES AND LEAD–IN CABLE AND SLIDE ON HEAT SHRINKS



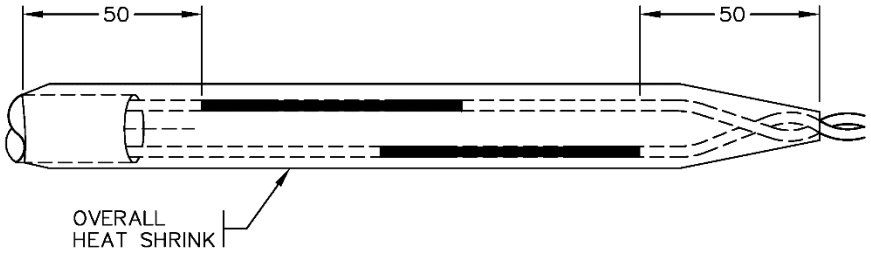
STEP 2. CONNECT AND SOLDER.
TWIST BARE CONDUCTORS TOGETHER AND SOLDER WITH 60/40 (TIN/LEAD) RESIN SOLDER.



STEP 3. INSULATE EACH SOLDER JOINT SEPARATELY



STEP 4. ENVIRONMENTALLY SEAL TOTAL SPLICE.



NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES.
3. HEAT SHRINK SHALL BE PREMIER CLEAR–SEAL (6mm I.D. INDIVIDUAL CONDUCTORS & 10mm I.D. FOR OVERALL) OR APPROVED ALTERNATE.

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

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C		
B	GENERAL REVISIONS	APR 17
A	DRAWING NUMBER CHANGED	NOV 98

DETECTOR LOOP TO SHIELDED CABLE SPLICING DETAILS

Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.8.7

STEP BY STEP LOOP INSTALLATION PROCEDURES AND RULES ARE AS FOLLOWS:

STEP 1 CONFIRM THE TYPE OF LOOP TO BE INSTALLED,
LOOP TYPES ARE DETAILED ON DRAWINGS SP635-2.8.2 to -2.8.4.

STEP 2 LAYOUT DETECTOR LOOPS AND REVIEW LOCATIONS WITH THE MINISTRY REPRESENTATIVE PRIOR TO SAWCUTTING THE ROADWAY. THE GENERAL LAYOUT OF THE DETECTOR LOOPS IS INDICATED ON DRAWINGS SP635-2.8.11 to -2.8.14. STOP BARS AND LANE LINES MUST BE LAID OUT PRIOR TO LOCATING DETECTOR LOOPS.

RULE 1 DETECTOR LOOPS SHALL NOT BE INSTALLED WHEN THE ROAD IS WET OR WHEN THE AMBIENT (AIR) TEMPERATURE IS LOWER THAN 5°C, UNLESS APPROVED IN WRITING BY THE MINISTRY ELECTRICAL REPRESENTATIVE. SEALANTS DO NOT ADHERE PROPERLY IN WET CONDITIONS. SHOULD THE CONTRACTOR BE ASKED IN WRITING BY THE MINISTRY ELECTRICAL REPRESENTATIVE TO INSTALL LOOPS IN THE WET AND/OR WHEN THE AIR TEMPERATURE IS BELOW 5°C, THE INSTALLATION WARRANTY WILL NOT BE ENFORCED.

RULE 2 DETECTOR LOOPS SHALL NOT BE INSTALLED WHEN THE PAVEMENT IS CRACKED OR BADLY RUTTED UNLESS THE INSTALLATION IS APPROVED IN WRITING BY THE MINISTRY ELECTRICAL REPRESENTATIVE. SAW CUTS CAN OFTEN CAUSE PAVEMENT CONDITIONS TO DETERIORATE FURTHER. IF RE-SURFACING OF THE INTERSECTION IS NOT PLANNED THEN PHOTOGRAPHS SHOULD BE TAKEN TO DOCUMENT THE PAVEMENT CONDITIONS BEFORE AND AFTER THE LOOP INSTALLATION. PHOTOGRAPHS SHALL BE LABELED WITH THE LOOP NUMBERS AND THEN SUBMITTED TO THE THE MINISTRY ELECTRICAL REPRESENTATIVE AFTER THE INSTALLATION IS COMPLETED.

STEP 3 CUT LOOP AND HOME RUN SLOTS IN ASPHALT.
ALL LOOP AND HOME RUN SLOTS SHALL BE CUT TO THE SAME DEPTH, WITH A PAVEMENT SAW. SLOTS SHALL NOT PASS THROUGH PAVEMENT INTO THE BASE GRAVEL.

RULE 3 LOOP AND HOME RUN SLOT MUST BE INSTALLED AT LEAST 300mm FROM ANY OTHER LOOP AND EACH LEAD-IN SLOTS, EXCEPT WHERE THE LEAD-IN CONDUCTORS ENTER THE 1" RPVC CONDUIT. THIS WILL REDUCE THE PROBABILITY OF INTERFERENCE BETWEEN LOOPS.

RULE 4 WHEN REPLACING LOOPS, CUT THROUGH TWICE ON EACH SIDE OF EXISTING LOOP. THIS MAY REQUIRE ADDITIONAL SAW CUTS, IF THE EXISTING LOOP IS NOT LOCATED IN THE SAW CUT PATH OF THE NEW LOOP. THIS WILL ELIMINATE THE POSSIBILITY OF INTERFERENCE BETWEEN THE OLD AND THE NEW LOOP.

STEP 4 CLEAN THE SLOT WITH A PROFESSIONAL GRADE PRESSURIZED WATER SYSTEM. REMOVE ALL WATER AND DIRT OUT OF THE SLOT CUT AND THE SURROUNDING 100mm OF ROAD SURFACE USING COMPRESSED AIR. SLOT MUST REMAIN COMPLETELY CLEAN AND DRY UNTIL THE SLOT IS SEALED.

STEP 5 INSTALL THE LOOP CONDUCTOR INTO THE LOOP SLOT. ENSURE CONDUCTORS ARE TIGHTLY WOUND AND PUSHED INTO THE BOTTOM OF THE SLOT. TWIST CONDUCTOR HOME RUN AT 15 TURNS PER METRE. INSTALL 75mm STRIPS OF BACKER ROD EVERY 600mm TO HOLD CONDUCTORS INTO SLOT.

RULE 5 ONLY ONE CONTINUOUS CONDUCTOR SHALL BE INSTALLED IN EACH LOOP AND HOME RUN SLOT TO THE JUNCTION BOX.

RULE 6 LOOP CONDUCTORS MUST BE INSTALLED IMMEDIATELY AFTER THE LOOP AND HOMERUN SLOTS ARE CUT.

STEP 6 INSTALL LOOP SEALANT AFTER CONDUCTORS HAVE BEEN INSTALLED. LOOP SEALANTS SHALL BE HEATED AS PER MANUFACTURER'S INSTRUCTIONS AND NEATLY APPLIED USING A FUNNEL WITH A NARROW SPOUT. ANY EXCESS SEALANT ON ROAD SURFACE SHALL BE REMOVED. AN ADDITIONAL APPLICATION OF LOOP SEALANT MAY BE REQUIRED WHERE THE SEALANT IS NOT UP TO THE PAVEMENT GRADE.

STEP 7 ONCE THE SEALING OF THE SLOT HAS BEEN PROPERLY COMPLETED, A DUST SUCH AS PORTLAND CEMENT SHALL BE SPRINKLED ONTO THE SEALANT TO PREVENT TRACKING BY ROADWAY TRAFFIC. ANY EXCESS DUST SHALL BE SWEPT OFF THE ROADWAY PRIOR TO ALLOWING TRAFFIC TO PASS OVER THE SEALED SLOT.

RULE 7 SPLICES WILL NOT BE ALLOWED IN LOOP CONDUCTORS OR SHIELDED CABLES.

NOTES
1. SEE DRAWING SP635-2.8.9 FOR CONTINUATION OF PROCEDURES AND RULES.

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
D	GENERAL REVISIONS	APR 20
C	GENERAL REVISIONS	APR 17
B	MINISTRY REPRESENTATIVE CHANGED TO MINISTRY ELECTRICAL REPRESENTATIVE	OCT 03
A	DRAWING NUMBER CHANGED, ROUND LOOPS ADDED	NOV 98

DETECTOR LOOP INSTALLATION PROCEDURES AND RULES

Date Approved
15/11/95 M.C. (Signature on File)
Chief Highway Engineer

SPECIFICATION DRAWING No.
SP635-2.8.8

CONTINUED FROM DRAWING SP635–2.8.8

STEP 8 A LOOP CHECK SHEET AS SHOWN ON DRAWING SP635–2.8.1 SHALL BE COMPLETED EACH TIME A NEW LOOP IS INSTALLED. PAVEMENT CONDITIONS ARE TO BE RECORDED BEFORE ANY SAW CUTS HAVE BEEN MADE. WEATHER CONDITIONS ARE TO BE RECORDED AT THE TIME THAT THE LOOPS ARE SEALED. MEASUREMENTS SHALL BE TAKEN AT THE JUNCTION BOX CLOSEST TO THE LOOP. (SEE RULE 8)

STEP 9 THE SPLICES BETWEEN DETECTOR LOOP CONDUCTORS AND THE SHIELDED CABLE ARE TO BE SOLDERED AND SEALED WITH HEAT SHRINK IN ACCORDANCE WITH DRAWING SP635–2.8.7.

STEP 10 REPEAT STEP 8 AT THE JUNCTION BOX OR VAULT NEAREST CONTROLLER.

RULE 8 LOOP DETECTOR RESISTANCE TO GROUND SHALL BE GREATER THAN 1 MEGAOHM, LOOP RESISTANCE SHALL BE FROM 0 TO 5 OHMS & LOOP INDUCTANCE SHALL BE WITHIN 20% OF THE VALUES SHOWN ON DRAWINGS SP635–2.8.2 TO –2.8.4 OR AS NOTED ON THE PLANS.

STEP 11 SUBMIT THE COMPLETED LOOP CHECK SHEET AND ANY RELEVANT PHOTOGRAPHS TO THE MINISTRY REPRESENTATIVE. THE TOTAL CIRCUIT INDUCTANCE VALUES AND THE MEGGER TEST VALUES WILL BE USED AS A BASELINE TO VERIFY DEGRADATION IN LOOP PERFORMANCE.

STEP 12 TAG EACH LOOP CABLE AS NOTED ON DRAWINGS SP635–2.5.5 AND –2.5.7.

RULE 9 MAINTAIN THE MAXIMUM SEPARATION POSSIBLE IN THE JUNCTION BETWEEN THE LOOP CONDUCTORS AND POWER CONDUCTORS.

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No.	Revision	Date
F		
E	STEP 11 UPDATED	APR 20
D	GENERAL REVISIONS	APR 17
C	STEP 11 CLARIFIED	OCT 03
B	DRAWING NUMBER CHANGED	NOV 98
A	RULE 8 REVISED	AUG 96

DETECTOR LOOP INSTALLATION PROCEDURES AND RULES

Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.8.9

REFER TO RULE 8 ON DRAWING SP635–2.8.9 FOR MINIMUM ALLOWABLE GROUND TO RESISTANCE AND MAXIMUM VARIATION IN INDUCTANCE VALUES.

BRITISH COLUMBIA
Ministry of Transportation and Infrastructure

Location: _____ Date: _____ Drawing No. _____

Electrician\Contractor: _____ Company: _____ Ministry Rep.: _____

Loop Sealant Used: _____ Pavement Conditions*: _____

Weather Conditions: Air Temp: _____ Precipitation: _____

Loop # as per DWG									
Phase assignment as per controller									
Resistance to ground** (ohms)	at loop								
	at controller								
Loop Resistance (ohms)	at loop								
	at controller								
Loop Inductance (micro Henrys)	at loop								
	at controller								

Loop # as per DWG									
Phase assignment as per controller									
Resistance to ground** (ohms)	at loop								
	at controller								
Loop Resistance (ohms)	at loop								
	at controller								
Loop Inductance (micro Henrys)	at loop								
	at controller								

* example of possible pavement conditions: good, cracked, sealed cracked, ruts at stop bar, pavement patches

** megger test – max 250V DC

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B	GENERAL REVISIONS	APR 17
A	DRAWING NUMBERS CHANGED	NOV 98

LOOP CHECK SHEET

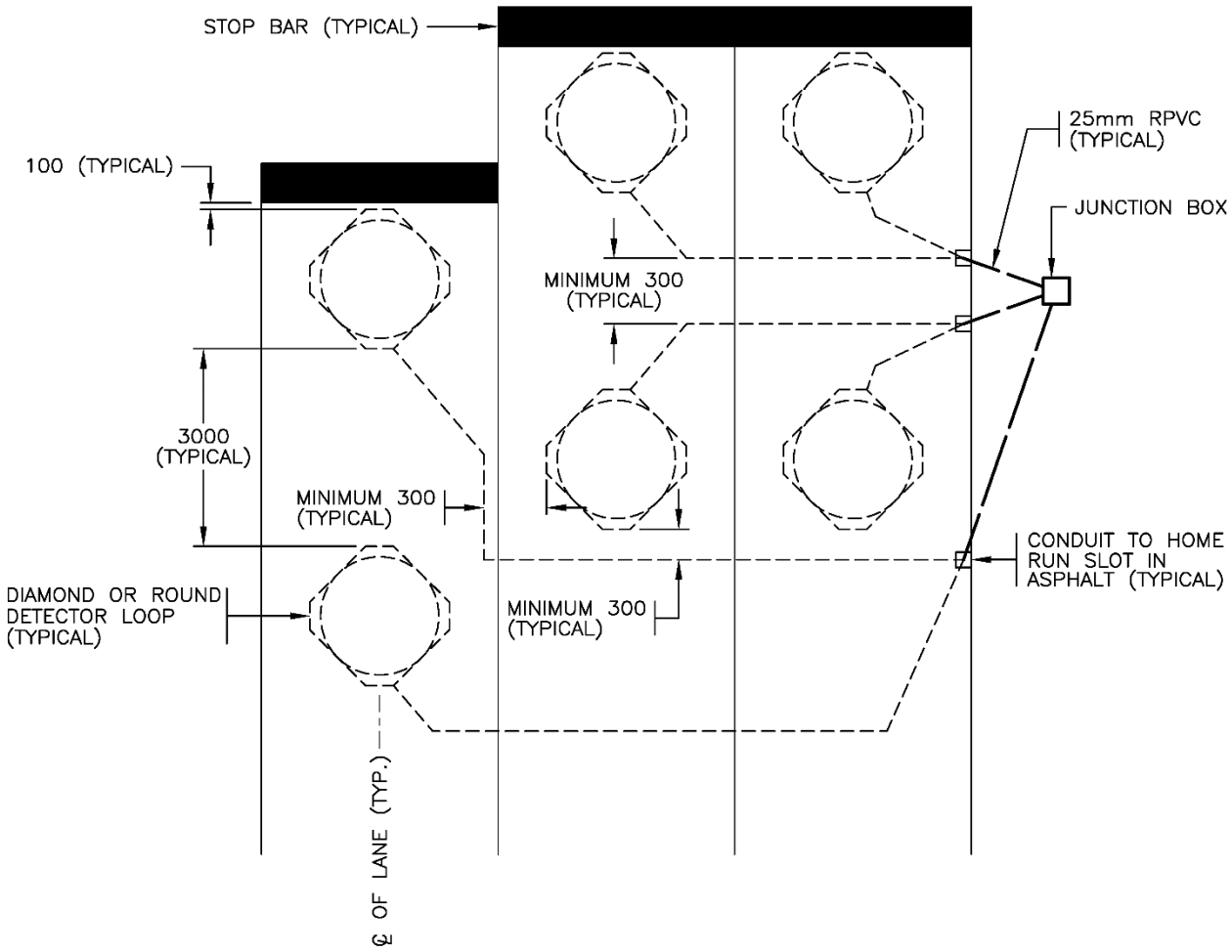
Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.8.10



TYPICAL LOOP LAYOUT

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. LOCATE LOOP CONDUITS IN ASPHALT TO SUIT LOOP LOCATIONS & MINIMIZE THE LENGTHS OF HOME RUN SLOTS IN THE ASPHALT.

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NOT TO SCALE

No.	Revision	Date
E		
D		
C		
B	GENERAL REVISIONS	APR 17
A	DRAWING NUMBER CHANGED, ROUND LOOPS ADDED	NOV 98

LAYOUT FOR DIAMOND OR ROUND TRAFFIC SIGNAL DETECTOR LOOPS

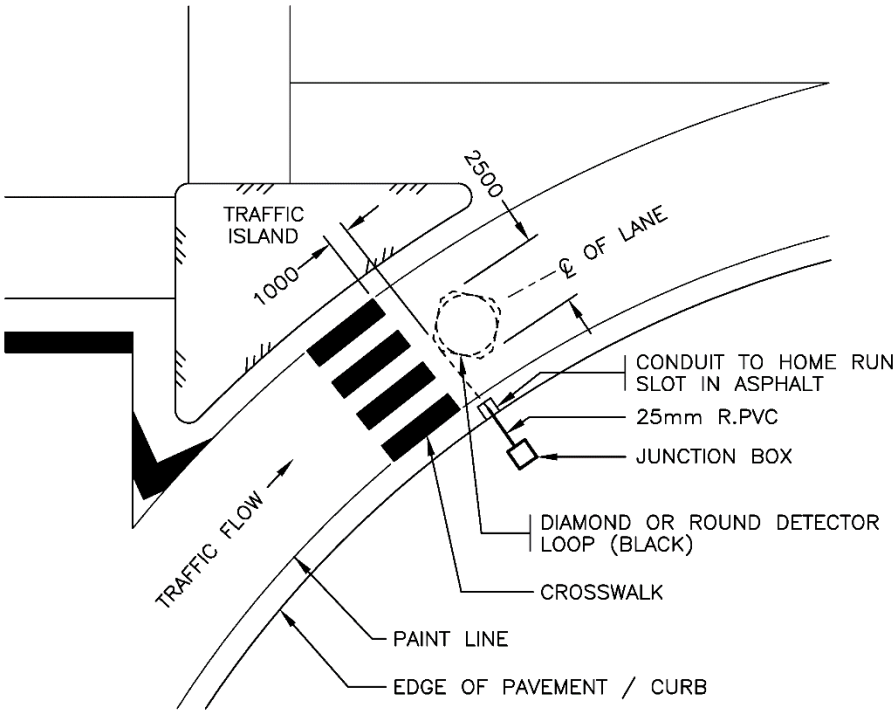
Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.8.11



RIGHT TURN LANE WITH TRAFFIC ISLAND SCENARIO SHOWN

PAVEMENT WIDTH UP TO 5900

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. FOR DIAMOND LOOP DETECTOR DETAILS SEE DRAWING SP635–2.8.2
   FOR ROUND LOOP DETECTOR DETAILS SEE DRAWING SP635–2.8.3
4. LOCATE LOOP CONDUITS IN ASPHALT TO SUIT LOOP LOCATIONS & MINIMIZE THE LENGTHS OF HOME RUN SLOTS IN THE ASPHALT.

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

NOT TO SCALE

No.	Revision	Date
E		
D		
C		
B	GENERAL REVISIONS	APR 17
A	DRAWING NUMBER CHANGED, ROUND LOOPS ADDED	NOV 98

LAYOUT FOR DIAMOND OR ROUND COUNTING DETECTOR LOOP

Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–2.8.12